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Targeted stimulation of motor cortex ensembles drives learned movements

https://doi.org/10.1038/s41586-026-11174-3

Received: 13 December 2025

Accepted: 16 September 2026

Published online: 14 October 2026

Check for updates

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During the execution of learned motor skills, the neural population in layer 2/3 (L2/3) of primary motor cortex (M1) expresses a reproducible spatiotemporal activity pattern[1]. It is debated whether M1 actively participates in generating this learned pattern and whether this pattern causally drives the learned movement[2–5]. Here we combined in vivo two-photon calcium imaging and holographic optogenetic stimulation of functionally defined M1 L2/3 ensembles in mice performing a skilled lever-pressing task. Brief synchronous stimulation of about 20 neurons whose activity onset precedes voluntary movement onset induced movements resembling the learned movement, while producing activity patterns in other M1 neurons resembling the learned activity. Moreover, trial-by-trial variability of optogenetically triggered population activity correlated with variability in the induced movements. These variabilities were predicted by the initial state of M1 activity preceding stimulation. In some trials, stimulation induced the learned activity without inducing movements, indicating that the learned activity is not simply a reflection of movements but can arise internally. Stimulation failed to generate movements or learned activity in task-disengaged mice. Stimulation of neurons whose activity followed movement onsets failed to induce the learned movement in task-engaged mice. Together, the learned activity in M1 L2/3 can be generated when M1 is prepared at the optimal initial state and receives precise triggering inputs, supporting the active role of M1 in generating learned activity and learned movements.

Learning and generating motor skills are core functions of the brain, enabling animals to engage with their environment in a purposeful and adaptive manner. The primary motor cortex (M1) is an important structure for these functions[6–8]. Motor learning induces multiple forms of plasticity in M1, particularly in layer 2/3 (L2/3) neurons, which undergo extensive synaptic reorganization[1,9–11]. This synaptic plasticity is accompanied by the emergence of a reproducible spatiotemporal pattern in the population activity of M1 L2/3 neurons during the execution of the learned motor skill[1]. Numerous studies have investigated the role of the learned activity pattern in M1 through neural recordings and loss-of-function experiments[2–5]. However, it remains an open question how the learned activity pattern is generated. Furthermore, whether the learned activity pattern causally instructs the learned movement pattern is unknown. We addressed these issues by combining in vivo two-photon imaging with two-photon holographic optogenetic stimulation of movement-related neuronal ensembles in M1 L2/3 in mice performing a forelimb motor skill.

Stimulation drives learned movement

We used a cued lever-press task, in which water-restricted mice were trained to grasp a lever and, in response to a sound cue delivered after a variable inter-trial interval, push and hold the lever for 1 s to acquire a water reward (Fig. 1a). After 2 weeks of daily training, mice became experts at this task, collecting rewards in most trials with short latencies (Extended Data Fig. 1a,b). Furthermore, the movement patterns became consistent across trials, a hallmark of learned motor skills (Extended Data Fig. 1b). These features of motor learning are consistent with previous reports using similar tasks[1,5,9–13]. It has been shown that the execution of the motor skill at this expert stage requires M1 activity[5]. It was also found that training with these tasks over additional months can make the behaviour independent of M1 (refs. 5,14), and so in this study we focused our experiments on the early expert stage (2–3 weeks).

To record and stimulate the M1 L2/3 population, we injected a bicistronic AAV vector to co-express the calcium indicator GCaMP6m and the excitatory opsin ChRmine in the same neurons[15]. We applied in vivo two-photon calcium imaging to record M1 L2/3 activity in task-performing mice (Fig. 1b and Extended Data Fig. 1c). This approach showed heterogeneous activity, including neurons showing significantly elevated activity ('movement-active', 30.2%) or decreased activity ('movement-suppressed', 16.4%) during movements (Extended Data Fig. 1d). The activity onset of individual movement-active neurons relative to movement onset tiled the duration of movements. Of these neurons, we identified a group of 'early-onset neurons' (12.1 ± 1.5% of all

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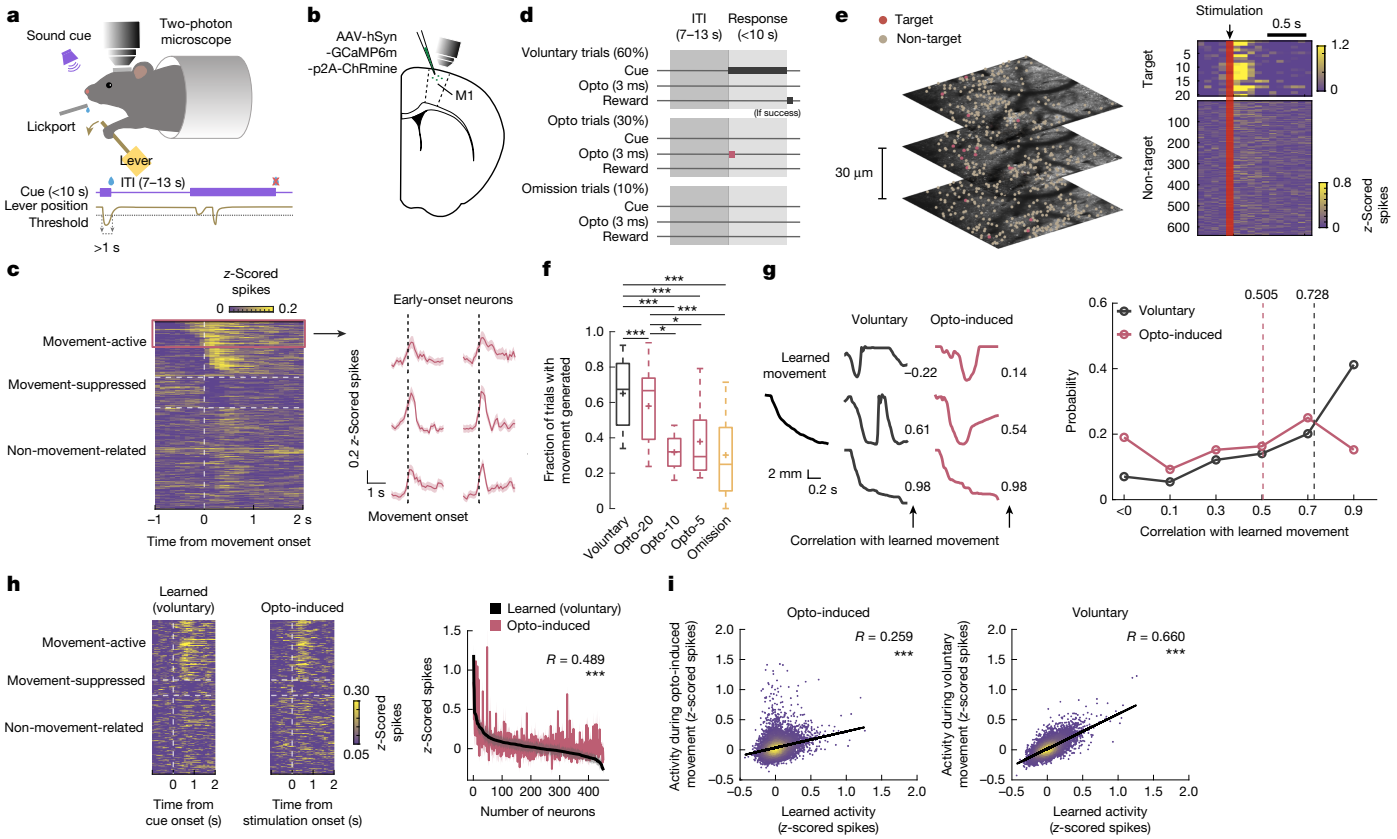


Fig. 1 | Holographic stimulation of early-onset neuronal ensemble drives learned movement. a, Schematic of experimental setup and task structure. ITI, inter-trial interval. **b**, Schematic of viral injection into M1 to label L2/3 neurons. **c**, Left, activity of movement-active (top, sorted by activity onset), movement-suppressed (middle) and non-movement-related (bottom) neurons averaged across movements and aligned to movement onset. The vertical dashed line indicates movement onset. Each row represents one neuron. $n = 13{,}874/25/15$ (neurons/sessions/mice). Right, activity of example early-onset movement-active neurons averaged across movements, aligned to movement onset. Mean ± SEM. **d**, Trial structures for holographic stimulation during behavioural sessions. **e**, Left, example two-photon calcium imaging of three planes in M1 L2/3 spanning 60 µm in depth, showing target neurons (red) and non-target neurons (beige). Right, trial-averaged responses of early-onset target and non-target neurons in opto trials aligned to stimulation onset. The red vertical bar indicates stimulation timing. **f**, Fraction of trials with a movement generated within 0.3 s from trial onset. Opto-20, -10 and -5 correspond to opto trials in which 20, 10, and 5 early-onset neurons were stimulated, respectively (voluntary compared with opto-20: $P = 8.39 \times 10^{-5}$, voluntary compared with opto-10: $P = 9.73 \times 10^{-7}$, voluntary compared with opto-5: $P = 6.47 \times 10^{-6}$, voluntary compared with omission: $P = 6.09 \times 10^{-26}$, opto-20 compared with opto-10: $P = 0.010$, opto-20 compared with opto-5: $P = 0.042$, opto-20 compared with omission: $P = 1.23 \times 10^{-12}$, opto-10 compared with opto-5: $P = 0.574$, opto-10 compared with omission: $P = 0.698$, opto-5 compared with omission: $P = 0.773$ for trial type, mixed-effects model, two-sided). Box and whisker plots: median and interquartile range; plus indicates mean. Voluntary: $n = 1{,}191/25/14$; opto-20: $n = 328/15/11$; opto-10: $n = 95/4/4$; opto-5: $n = 140/6/4$; omission: $n = 201/25/14$ (trials/sessions/mice). **g**, Left, example movements from one session. The learned movement, voluntary movements and opto-induced movements of various correlations with the learned movement are shown. Right, probability of movements with varying correlations with the learned movement in voluntary and opto trials. The dashed lines indicate the medians (voluntary: 0.728, opto: 0.505). Voluntary: $n = 313/15/11$; opto: $n = 184/15/11$ (trials/sessions/mice). **h**, Left, activity of movement-active, movement-suppressed and non-movement-related neurons (separated by horizontal dashed lines) averaged across learned and opto-induced movements separately, from an example session. Each row represents a non-target neuron. Right, average activity of non-target neurons during the 1 s following cue/stimulation onset, sorted by amplitudes in learned movements from the same example session (Pearson's correlation, two-sided; $P = 1.54 \times 10^{-28}$, $R = 0.489$). Mean ± SEM. **i**, Left, average activity of non-target neurons during voluntary learned movements compared with those during opto-induced movements pooled across sessions ($P = 9.93 \times 10^{-135}$ for learned activity, mixed-effects model; Pearson's correlation, $R = 0.259$). Each dot represents the activity of one neuron averaged during the 1 s following cue/stimulation onset, averaged across trials. Right, average activity of non-target neurons during voluntary learned movements compared with those during all voluntary movements pooled across sessions ($P < 1.00 \times 10^{-308}$ for learned activity, mixed-effects model; Pearson's correlation, $R = 0.660$). Each dot represents the activity of one neuron averaged during the 1 s following cue onset, averaged across trials. $n = 8{,}789/15/11$ (neurons/sessions/mice). For this and all other figures, $^*P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$.

neurons per session, mean ± standard error of the mean (SEM)) whose activity onset preceded the movement onset (Fig. 1c).

We proposed that activity in these early-onset neurons contributed to the initiation of the learned movement. To test this hypothesis, we leveraged two-photon holographic optogenetic stimulation with a spatial light modulator that enables the activation of arbitrary groups of neurons within the field of view with single-cell resolution[15–19]. We used this approach to ask whether an artificial stimulation of the early-onset neurons could induce the learned movement. We first performed two-photon calcium imaging for about 60 trials, imaging at three depths within M1 L2/3 semi-simultaneously, each spaced 30 µm apart, to capture 555 ± 42 neurons (mean ± SEM). The data were quickly analysed offline to identify early-onset neurons, after which the same imaging fields were identified, and the behavioural session was resumed. From this point forward, most of the trials were normal voluntary trials in which a lever press after the sound cue induced a water reward. In 30% of randomly interspersed trials (opto trials), the sound cue was omitted, and instead a brief 3 ms pulse of optogenetic

stimulation was delivered to either 5, 10, or 20 early-onset neurons simultaneously at the trial onset (Methods and Extended Data Fig. 1e). In opto trials, no reward was given even when mice made a successful movement. We also included omission trials, in which the sound cue and optogenetic stimulation were omitted, and no reward was ever given (Fig. 1d). In the opto trials, the stimulation successfully elicited reliable activity in the target neurons as revealed by simultaneous calcium imaging (Fig. 1e).

We observed that the stimulation of early-onset neurons can trigger movements. We focused our analysis on lever movements initiated within a short latency (0.3 s) after trial onset (Methods and Extended Data Fig. 2). Although the fraction of trials in which movements were initiated was the highest in voluntary trials, fractions in 20-neuron opto trials were substantially higher than in omission trials (Supplementary Table 1). This indicates that movements in 20-neuron opto trials were driven by the stimulation as opposed to voluntary movements serendipitously occurring after the stimulation (Fig. 1f). By contrast, 5- and 10-neuron opto trials did not induce movements more frequently than in omission trials, indicating that these were not sufficient stimulation to reliably induce movements. Consequently, we focused the following experiments and analysis on the 20-neuron model. We next examined whether the opto-induced movements resembled the learned movement, which was defined for each session by identifying the subset of the voluntary movements that exhibited high stereotypy and averaging the movements in a random half of those trials. The trials used to construct the learned movement were excluded from subsequent analysis to avoid self-comparisons (Methods). Both voluntary movements and opto-induced movements in individual trials showed variable correlations with the learned movement, with voluntary movements showing an overall higher correlation. Notably, however, about half of the opto-induced movements showed a high (>0.5) correlation with the learned movement (Fig. 1g). Thus, a brief and synchronous stimulation of about 20 early-onset neurons frequently induced movements, many but not all of which resembled the learned movement.

The question arises as to how 20 neurons could drive the learned movement. M1 L2/3 is a highly recurrent network, in which excitatory neurons form dense interconnections[20–23]. We proposed that the activation of these approximately 20 neurons elicits population activity in the other, non-target neurons that resembles the activity during the voluntary learned movement. To investigate this, we analysed the trial-averaged activity of the non-target neurons during the voluntary trials in which the animal generated the learned movement ('learned activity') and compared that with the trial-averaged activity during opto-induced movements. We found that the activity of non-target neurons during opto-induced movements correlated with that during the voluntary learned movements (Fig. 1h,i). In other words, neurons that were activated and suppressed during the voluntary learned movements tended to be also activated and suppressed during opto-induced movements, respectively. To explore the upper bound of this correlation, we repeated this analysis using the activity during voluntary trials (note that not all voluntary trials resulted in the learned movement; Fig. 1g). The voluntary trial activity correlated with the learned activity, similarly to (but more strongly than) the opto-induced activity (Fig. 1i). Thus, our stimulation of about 20 early-onset neurons induced population activity patterns that resembled the learned pattern in the non-target neurons that are not directly stimulated.

Learned movement requires specific input

It is striking that our artificial stimulation of a small group of neurons induced naturalistic population activity. One possibility is that any random input can engage the local pattern-completion network and induce the same naturalistic activity to drive movements. To test this, we conducted another set of experiments, this time targeting a subset of 'late-onset neurons' (15.9 ± 0.9% of all neurons per session, mean ± SEM), whose activity onsets were after movement onset in voluntary movements (Fig. 2a and Extended Data Fig. 3). About 20 of the late-onset neurons were randomly selected as target neurons, matching the number stimulated for the early-target stimulation experiments. The stimulation was done during behavioural sessions in which most of the trials were voluntary trials, in the same way as early-target stimulation sessions.

Late-target stimulation induced movements in a subset of stimulation trials, showing a non-significant trend towards lower efficacy compared with voluntary and early-target trials (Fig. 2b). We next examined the quality of stimulation-induced movements by evaluating the correlation with the learned movement. We found that the movements induced by late-target stimulation were significantly less similar to the learned pattern than the movements in voluntary and early-target stimulation trials (Fig. 2c). Thus, compared with early-target stimulation, late-target stimulation did not reliably induce the learned movement. These results indicate that the choice of target neurons is important—early-target stimulation is much more effective at inducing the learned movement than late-target stimulation.

We also observed differences in the reaction time of movement initiation across trial types. Specifically, opto-induced movements in both early-target and late-target stimulation trials had a significantly shorter reaction time than voluntary movements (Fig. 2d). The shorter reaction time supports the idea that holographic stimulation of M1 L2/3 neurons bypassed the sensory-motor transformation taking place in voluntary trials, leading to faster movement onsets. The reaction time was shorter in late-target stimulation than in early-target stimulation. This raises the possibility that early-target stimulation induced an additional circuit computation to perform pattern completion to generate the learned activity and movement.

We also examined other movement parameters across trial types, including average movement speed and total lever displacement (Extended Data Fig. 4a,b). Voluntary movements exhibited the highest average speed and greatest displacement. By contrast, these measures did not differ significantly among the remaining trial types, suggesting that the primary distinctions between conditions relate to their alignment with the learned movement pattern and reaction time, rather than overall movement magnitude or speed.

We next explored the neural underpinnings that differentiate behavioural outcomes between early-target and late-target stimulation. To this end, we analysed the activity of the non-target neurons within M1 L2/3 and how they were recruited by the stimulation of the target neurons. We found that early-target stimulation led to the activation of a larger fraction of non-target neurons, and higher response amplitudes of the recruited non-target neurons compared with late-target stimulation (Fig. 2e). Furthermore, we found that the population activity induced by late-target stimulation did not resemble the learned activity as much as early-target stimulation (Fig. 2f). Overall, stimulation of size-matched late-onset ensembles failed to reliably induce the learned movement, probably because the elicited population activity patterns did not sufficiently match the activity pattern underlying the learned movement. These findings suggest that generating the learned movement requires the activation of early-onset neuronal ensembles as the triggering input to the M1 network, which then propagate the activity to other nearby neurons to complete the learned activity pattern.

Although late-target stimulation was less reliable at inducing learned movements, it nevertheless occasionally did so. One possibility is that late-target stimulation sometimes engages the same movement initiation circuitry by recruiting early-onset neurons. We therefore examined how non-target early-onset neurons were recruited by target neuron stimulation across three conditions: early-target stimulation movement trials, late-target stimulation movement trials and late-target stimulation no-movement trials (Extended Data Fig. 5). We found that early-onset neurons are recruited during late-target stimulation, particularly in movement trials. However, the fraction

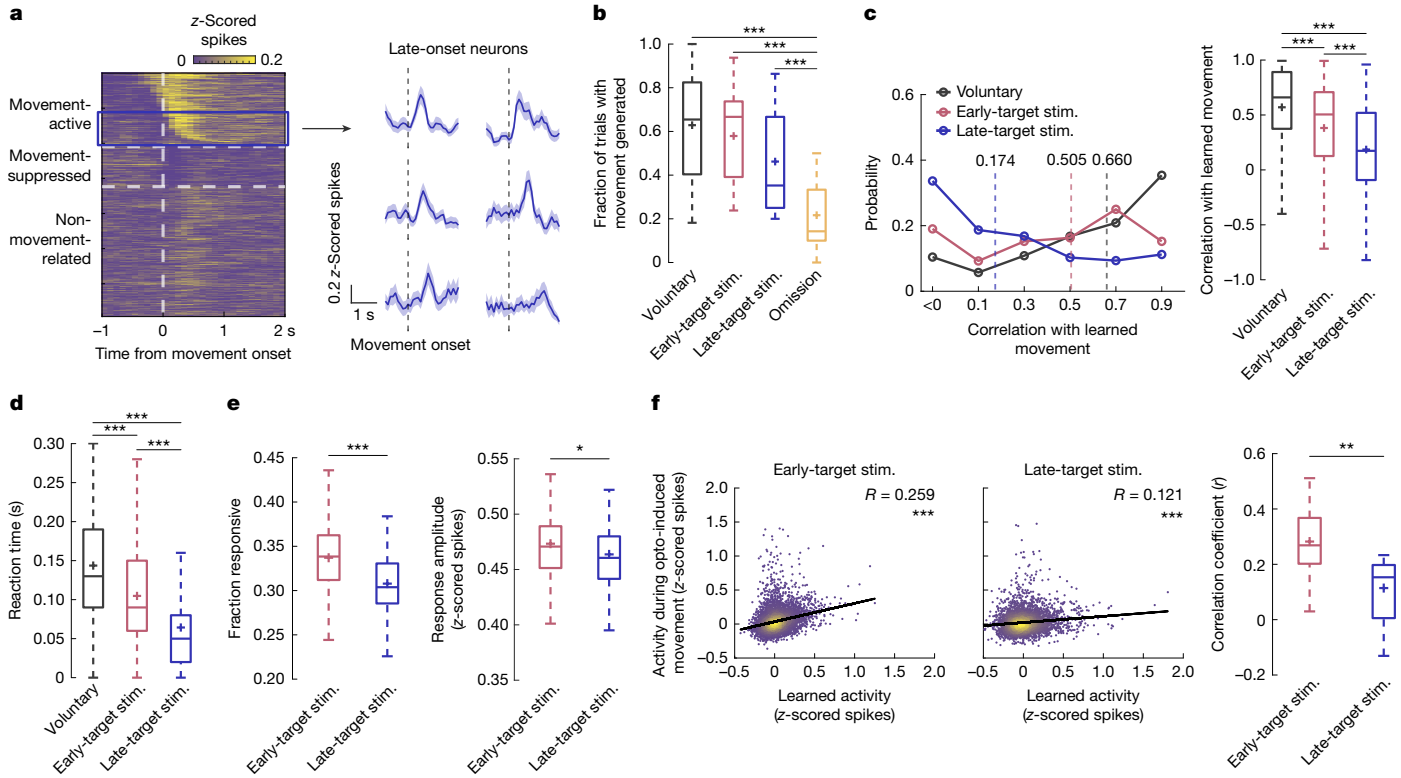


Fig. 2 | Stimulation of late-onset neuronal ensemble fails to drive learned movement. a, Left, activity of movement-active (top, sorted by activity onset), movement-suppressed (middle) and non-movement-related (bottom) neurons averaged across movements, and aligned to movement onset. The vertical dashed line indicates movement onset. Each row represents one neuron. n = 13,874/25/15 (neurons/sessions/mice). Same plot as Fig. 1c (left), repeated here to provide contextual reference for late-onset neurons. Right, activity of example late-onset movement-active neurons averaged across movements, aligned to movement onset. Mean ± SEM. **b**, Fraction of trials with a movement generated within 0.3 s from trial onset (voluntary compared with early: $P = 4.85 \times 10^{-5}$, voluntary compared with late: $P = 0.095$, voluntary compared with omission: $P = 6.93 \times 10^{-35}$, early compared with late: $P = 0.179$, early compared with omission: $P = 7.93 \times 10^{-18}$, late compared with omission: $P = 1.93 \times 10^{-7}$ for trial type, mixed-effects model, two-sided). Voluntary: n = 1,199/25/15; early: n = 328/15/11; late: n = 237/10/8; omission: n = 204/25/15 (trials/sessions/mice). stim., stimulation. **c**, Left, probability of movements with varying correlations with the learned movement for each trial type. Dashed lines indicate the medians (voluntary: 0.660, early: 0.505, late: 0.174). Right, correlation with the learned movement in each trial type (voluntary compared with early: $P = 5.24 \times 10^{-12}$, voluntary compared with late: $P = 1.23 \times 10^{-12}$, early compared with late: $P = 2.45 \times 10^{-4}$ for trial type, mixed-effects model, two-sided). Voluntary: n = 489/25/15; early: n = 184/15/11; late: n = 107/10/8 (trials/sessions/mice). **d**, Reaction time of movements in each trial type (voluntary compared with early: $P = 1.10 \times 10^{-11}$, voluntary compared with late: $P = 1.55 \times 10^{-30}$, early compared with late: $P = 1.98 \times 10^{-6}$ for trial type, mixed-effects model, two-sided). Voluntary: n = 729/25/15; early: n = 184/15/11; late: n = 107/10/8 (trials/sessions/mice). **e**, Left, fraction of responsive non-target neurons per trial for early-target and late-target stimulation trials ($P = 5.82 \times 10^{-5}$ for trial type, mixed-effects model, two-sided). Right, response amplitudes of the responsive non-target neurons for early-target and late-target stimulation trials (P = 0.018 for trial type, mixed-effects model, two-sided). Early: n = 184/15/11; late: n = 107/10/8 (trials/sessions/mice). **f**, Left, average activity of non-target neurons during voluntary learned movements compared with early-target stimulation trials pooled across sessions ($P = 9.93 \times 10^{-135}$ for learned activity, mixed-effects model; Pearson's correlation, R = 0.259). Each dot represents the activity of one neuron averaged during the 1 s following cue/stimulation onset, averaged across trials. Middle, average activity of non-target neurons during voluntary learned movements versus in late-target stimulation trials pooled across sessions ($P = 7.86 \times 10^{-20}$ for learned activity, mixed-effects model; Pearson's correlation, R = 0.121). Each dot represents the activity of one neuron averaged during the 1 s following cue/stimulation onset, averaged across trials. Right, correlation coefficients between non-target neuron population activity in voluntary learned movements and opto-induced movements calculated for individual sessions for early-target and late-target stimulation sessions, respectively ($P = 2.1 \times 10^{-3}$, Wilcoxon rank-sum test, two-sided). Early: n = 8,789/15/11; late: n = 5,252/10/8 (neurons/sessions/mice). For this figure, box-and-whisker plots: median and interquartile range; plus indicates mean.

of recruited neurons in late-target movement trials remained substantially lower than that observed during early-target stimulation movement trials.

Given the distinct behavioural and neural outcomes between early-target and late-target stimulation, we next asked whether these effects could be attributed to confounding differences in the properties of the target neurons. M1 neurons exhibit distinct input–output connectivity, somatotopic organization and functional specification depending on their anatomical location. To account for this, we assessed the depth, anterior–posterior and medial–lateral distributions of early and late targets, and confirmed that they were not significantly different from each other (Extended Data Fig. 6a). The degree of spatial clustering among neurons may reflect distinct local circuit organizations, in which stimulation of a clustered ensemble could be more effective at generating functional output[24]. However, we found that early and late targets exhibited comparable degrees of spatial clustering, which were similar to the overall M1 L2/3 population revealed by shuffling labels (Extended Data Fig. 6b). Other than their activity onset timing that differs by definition, their activity properties during voluntary movements were similar, as quantified by the peak amplitude and response consistency (Extended Data Fig. 6c). Collectively, these other properties of early- and late-onset target neurons are unlikely to explain the distinct outcomes of their stimulation.

Up to this point, we have focused our analysis on movement trials to study the differences across trial types when movements were induced. However, excluding the no-movement trials could introduce a bias.

We therefore repeated the population activity analyses, while including both movement and non-movement trials, including omission trials (Extended Data Fig. 7). We observed differences in population activity between early- and late-target trials, which were similar to the movement-trial analysis. Importantly, population activity during omission trials was markedly distinct from both photostimulation-evoked and learned activity patterns.

M1 explains movement variability

Opto-induced movements exhibited varying degrees of similarity to the learned movement pattern across trials. We next examined whether this variability is reflected in the variability of induced population activity. We focused on the early-target stimulation sessions, in which we successfully triggered movements that resembled the learned movement in a considerable fraction of trials. We applied principal component analysis (PCA) on the non-target population activity (voluntary trials only) and then projected the population activity of individual trials onto the neural space that captured 80% of the variance (Methods). In voluntary trials, on a trial-by-trial basis, there was a negative correlation between the population activity distance to the learned activity and movement correlation with the learned movement (Fig. 3a). In other words, the more similar the population activity was to the learned activity, the more similar the movement was to the learned movement. A similar result was observed in opto trials, in which movements with low correlations with the learned movement showed population activity trajectories distinct from the learned activity (Fig. 3b). These results were not affected when we included the target neurons in the analysis (Extended Data Fig. 8a). These findings suggest that the learned movement was generated when the induced population activity resembled the learned activity.

Apart from the similarity to the learned movement, a previous study showed that trial-by-trial variability in movement kinematics is represented in M1 L2/3 activity of trained animals[1]. Thus, we next examined the activity–movement relationship in pairs of trials to ask whether the relationship is shared across both the voluntary and opto-induced movements. Specifically, we examined voluntary trial pairs, opto trial pairs and voluntary–opto trial pairs to assess whether the movement similarity and the population activity similarity correlate with each other across voluntary and opto-induced movements. In all three comparisons, we observed a significant negative correlation between population activity distance and movement correlation (Fig. 3c). This suggests that M1 L2/3 activity reflects movement kinematics on a trial-by-trial basis, regardless of whether the movement was generated voluntarily or by our optogenetic stimulation.

We further explored the activity–movement relationship using a decoding analysis to decode the movement time series on individual trials with population activity using lasso regression. The decoder was trained using 80% of voluntary trials and tested on the other trials (Fig. 3d (left) and Methods). The decoder was able to decode the voluntary movements on the held-out trials substantially better than the chance level defined by trial shuffles (Fig. 3d, right). This confirms that M1 L2/3 activity contains information about movement kinematics. When the same decoder trained on voluntary trials was tested on opto trials, we again saw that it was able to decode movements substantially better than chance (Fig. 3d, right). This suggests that the mechanism by which M1 L2/3 drives movements was shared between voluntary and opto trials. The decoder performance was significantly better in opto trials than in held-out voluntary trials, although the decoder was trained with voluntary trials. This may suggest that voluntary movements are also influenced by additional sources of variability outside of M1, which dilutes the coupling between M1 activity and movement, whereas opto movements are more purely controlled by M1 activity.

Taken together, these findings indicate that variability in M1 L2/3 activity can explain variability of both voluntary and opto-induced movements.

Initial M1 state predicts movement

The next question is what accounts for variability in non-target population activity in opto trials, when the input was the same in every trial (stimulation of the same about 20 early-onset neurons). Previous studies have found that the variability in the state of motor cortex population activity before movement initiation contributes to movement variability[25–29]. Thus, we proposed that the initial population state before the stimulation influences the subsequent evolution of population activity, thereby contributing to movement variability. To address this hypothesis, we examined the population activity during the period (0.5 s) immediately before our stimulation in opto trials or cue onset in voluntary trials. In opto trials, we observed that the pre-stimulation population activity that was closer to the pre-cue state of the learned activity was associated with subsequent, opto-induced movements more similar to the learned movement (Fig. 4a and Extended Data Fig. 9a). In other words, the pre-stimulation activity state predicted whether the subsequent stimulation was going to be successful in inducing the learned movement. Similar results were found in voluntary trials, such that the pre-cue population state predicted the quality of the voluntary movements (Fig. 4b). These results were consistent whether or not the target neurons were included in the analysis (Extended Data Fig. 8b). Thus, in both the opto and voluntary trials, on a trial-by-trial basis, there seems to be an optimal initial state that facilitated the subsequent generation of the learned movement. This is consistent with the 'initial condition hypothesis', which posits that the initial state of a dynamical system partially defines the evolution of the subsequent spatiotemporal activity[29]. Supporting this notion, the pre-stimulation and pre-cue activity state predicted the subsequent population activity during movements on a trial-by-trial basis, in both the opto and voluntary trials, respectively (Fig. 4c and Extended Data Fig. 9b). Perhaps as a consequence, we found a negative correlation between the distance of initial states and movement correlation—in pairs of voluntary trials, pairs of opto trials and pairs of voluntary and opto trials (Fig. 4d and Extended Data Fig. 9c). Furthermore, the initial state significantly predicted the probability of movement in both opto and omission trials, but the relationship between initial state and movement probability differed significantly between the two conditions. Thus, the change in movement probability produced by photostimulation depended on the initial network state, directly demonstrating that the initial state constrains the effect of photostimulation (Methods and Extended Data Fig. 9d).

Taken together, the initial state of the M1 L2/3 population predicts the variability in subsequent M1 activity and movements.

Evoked activity is not movement-driven

When optogenetic stimulation was delivered to about 20 early-onset neurons, we observed activity in the non-target M1 L2/3 population that correlated with movement variability. This could be because the activity of target neurons propagates to other L2/3 neurons to generate the population activity. However, an alternative possibility is that the target neurons directly engage downstream circuits to generate movements, largely bypassing the local L2/3 network. In this scenario, M1 L2/3 population activity may reflect sensory feedback signals related to the resulting movement. To parse these possibilities, we leveraged the subset of trials in which early- or late-target stimulation failed to evoke any overt movements (Fig. 2c), which provided a window into network dynamics in the absence of motor output.

For both early- and late-target stimulation, the non-target neurons exhibited substantial activity in no-movement trials, indicating

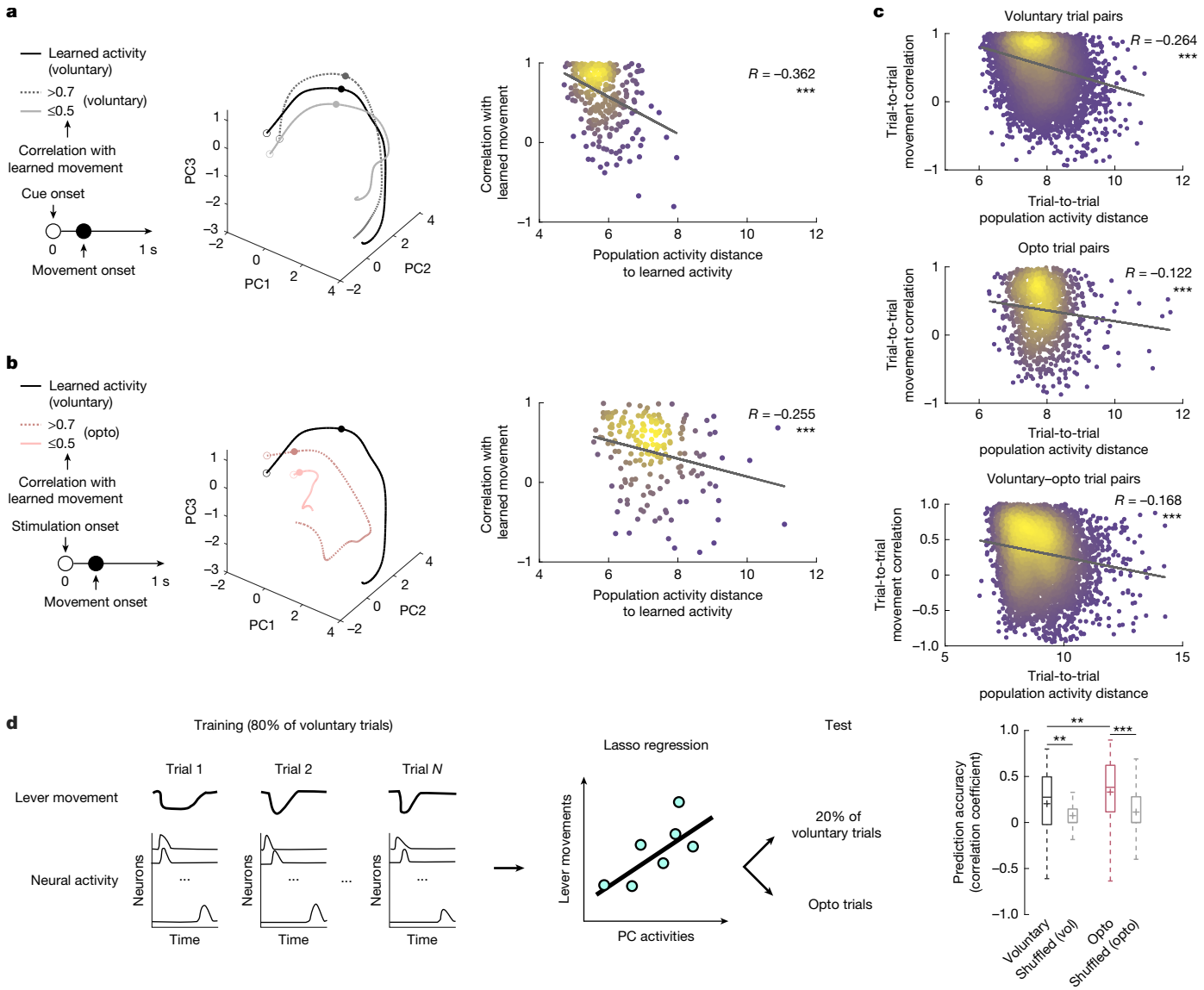


Fig. 3 | M1 activity explains movement variability. a, Left, population activity trajectories of all non-target neurons in voluntary trials of an example session, for the 1 s after cue onset. Black, learned activity; grey dotted and solid lines, trials with high (>0.7) and low (≤0.5) correlation with the learned movement, respectively. Right, population activity distance to the learned activity was negatively correlated with correlation with the learned movement ($P = 3.32 \times 10^{-11}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.362$). Each dot represents an individual trial. $n = 313/15/11$ (pairs/sessions/mice). **b**, Left, population activity trajectories of all non-target neurons in opto trials of an example session, for the 1 s period after stimulation onset. Black, learned activity; pink dotted and solid lines, trials with high (>0.7) and low (≤0.5) correlation to the learned movement, respectively. Right, population activity distance to the learned activity negatively correlated with correlation with the learned movement ($P = 4.53 \times 10^{-4}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.255$). Each dot represents an individual trial. $n = 184/15/11$ (pairs/sessions/mice). **c**, Population activity distance negatively correlated with movement correlation between voluntary trials (top, $P = 5.90 \times 10^{-90}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.264$), between opto trials (middle, $P = 5.63 \times 10^{-5}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.122$), and between voluntary and opto trials (bottom, $P = 1.77 \times 10^{-24}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.168$). Voluntary: $n = 8,377/15/11$; opto: $n = 1,173/15/11$; voluntary–opto: $n = 6,488/15/11$ (pairs/sessions/mice). **d**, Left, schematic illustrating the prediction of lever movements on individual trials based on M1 population activity. A regression model was trained using 80% of voluntary trials and tested on the remaining voluntary trials as well as opto trials (Methods). Right, prediction accuracy, measured by correlation coefficients between the actual and predicted lever movements, was significantly better than chance in both voluntary and opto trials (voluntary compared with shuffled: $P = 5.38 \times 10^{-3}$, opto compared with shuffled: $P = 5.84 \times 10^{-13}$, mixed-effects model), and better in opto trials compared with voluntary trials ($P = 1.39 \times 10^{-3}$ for trial type, mixed-effects model, two-sided). Box-and-whisker plots: median (voluntary: 0.275, voluntary shuffled: 3.56×10^{-17}, opto: 0.383, opto shuffled: 6.41×10^{-17}) and interquartile range; plus indicates mean. Voluntary: $n = 98/15/11$; shuffled (voluntary): $n = 98/15/11$; opto: $n = 184/15/11$; shuffled (opto): $n = 184/15/11$ (trials/sessions/mice).

that stimulation of 20 neurons can generate population activity in non-target neurons without engaging movements, although this activity tended to be less prolonged than in movement trials (Extended Data Fig. 10). We analysed this short-latency (0.3 s) population activity in terms of both the activity amplitude and correlation with the learned activity pattern. In early-target stimulation, both the amplitude and correlation of population activity were similar in movement and no-movement trials (Fig. 5a). Comparing non-target population activity between early- and late-target stimulation, we found that early-target stimulation evoked stronger activity that better resembled the learned activity than late-target stimulation in both movement and no-movement trials (Fig. 5b). These observations indicate

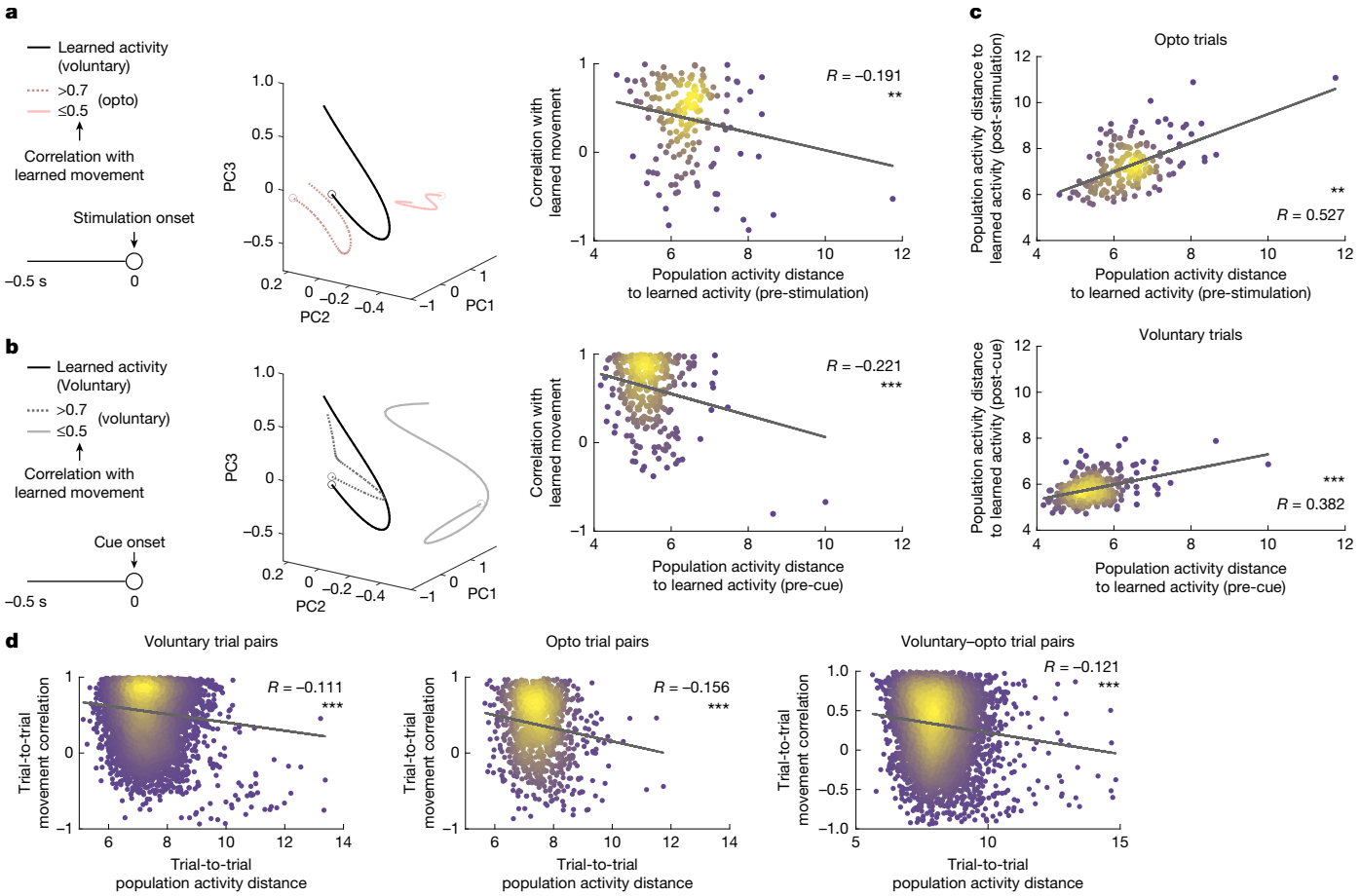


Fig. 4 | Initial state of M1 activity predicts movement variability. a, Left, population activity trajectories of all non-target neurons in opto trials of an example session, for the 0.5 s period before stimulation onset. Black, learned activity; pink dotted and solid lines, trials with high (>0.7) and low (≤0.5) correlation to the learned movement, respectively. Right, population activity distance to the learned activity negatively correlated with correlation with the learned movement ($P = 0.009$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.191$). Each dot represents an individual trial. **b**, Left, population activity trajectories of all non-target neurons in voluntary trials of an example session, for the 0.5 s period before cue onset. Black, learned activity; grey dotted and solid lines, trials with high (>0.7) and low (≤0.5) correlation with the learned movement, respectively. Right, population activity distance to the learned activity negatively correlated with correlation with the learned movement ($P = 7.68 \times 10^{-5}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.221$). Each dot represents an individual trial. **c**, Population activity distance to the learned activity showed a correlation between pre- and post-stimulation periods in opto trials (top, $P = 0.002$ for pre-stimulation activity distance, mixed-effects model; Pearson's correlation, $R = 0.527$) and between pre- and post-cue periods in voluntary trials (bottom, $P = 1.08 \times 10^{-8}$ for pre-cue activity distance, mixed-effects model; Pearson's correlation, $R = 0.382$). **d**, Population activity distance before the cue or stimulation negatively correlated with correlation of subsequent movements between voluntary trials (left, $P = 7.65 \times 10^{-14}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.111$), between opto trials (middle, $P = 3.00 \times 10^{-4}$, mixed-effects model, activity distance; Pearson's correlation, $R = -0.156$), and between voluntary and opto trials (right, $P = 1.16 \times 10^{-13}$ for activity distance, mixed-effects model; Pearson's correlation, $R = -0.121$). For **a**–**c**, opto: $n = 184/15/11$; voluntary: $n = 313/15/11$. For **d**, voluntary: $n = 8{,}377/15/11$; opto: $n = 1173/15/11$; voluntary–opto: 6,488/15/11 (pairs/sessions/mice).

that the distinct neural responses in early- and late-target stimulation arise from their differences in how they engage the local circuit, instead of as a consequence of the differences in the induced movements. Furthermore, they suggest that the learned activity induced by early-target stimulation is not purely a consequence of induced movements (for example, sensory feedback) and is rather completed internally in the brain.

Task engagement is required

We have shown that optogenetic stimulation of early-onset targets can induce the learned activity pattern and learned movement. Furthermore, generation of learned activity and movement depended on the initial state of M1. All the experiments so far were done in task-engaged animals that were presumably preparing the learned movement. We asked whether this behavioural context is important for the observed effects of our stimulation. To address this, we performed stimulation in a no-task condition in which well-trained mice were presented with the lever but received no sound cue and had no access to the lickport. We then delivered the same optogenetic stimulation used in the task context—interleaving early-target and late-target stimulation—each followed by a variable interval (Fig. 5c). In this no-task context, both early- and late-target stimulation failed to evoke any movement reliably (Fig. 5c), demonstrating that task engagement is important for the stimulation to produce motor outputs. We then examined non-target population activity following stimulation, comparing task and no-task contexts specifically in no-movement trials. Task engagement bidirectionally modulated the non-target response: early-target stimulation produced stronger activity in the task context, whereas late-target stimulation elicited weaker activity compared with the no-task condition. Moreover, the population activity evoked by stimulation in the no-task context did not resemble the learned activity pattern (Fig. 5d). Taken together, these findings indicate that stimulation of early-onset neurons drives movements in a context-dependent

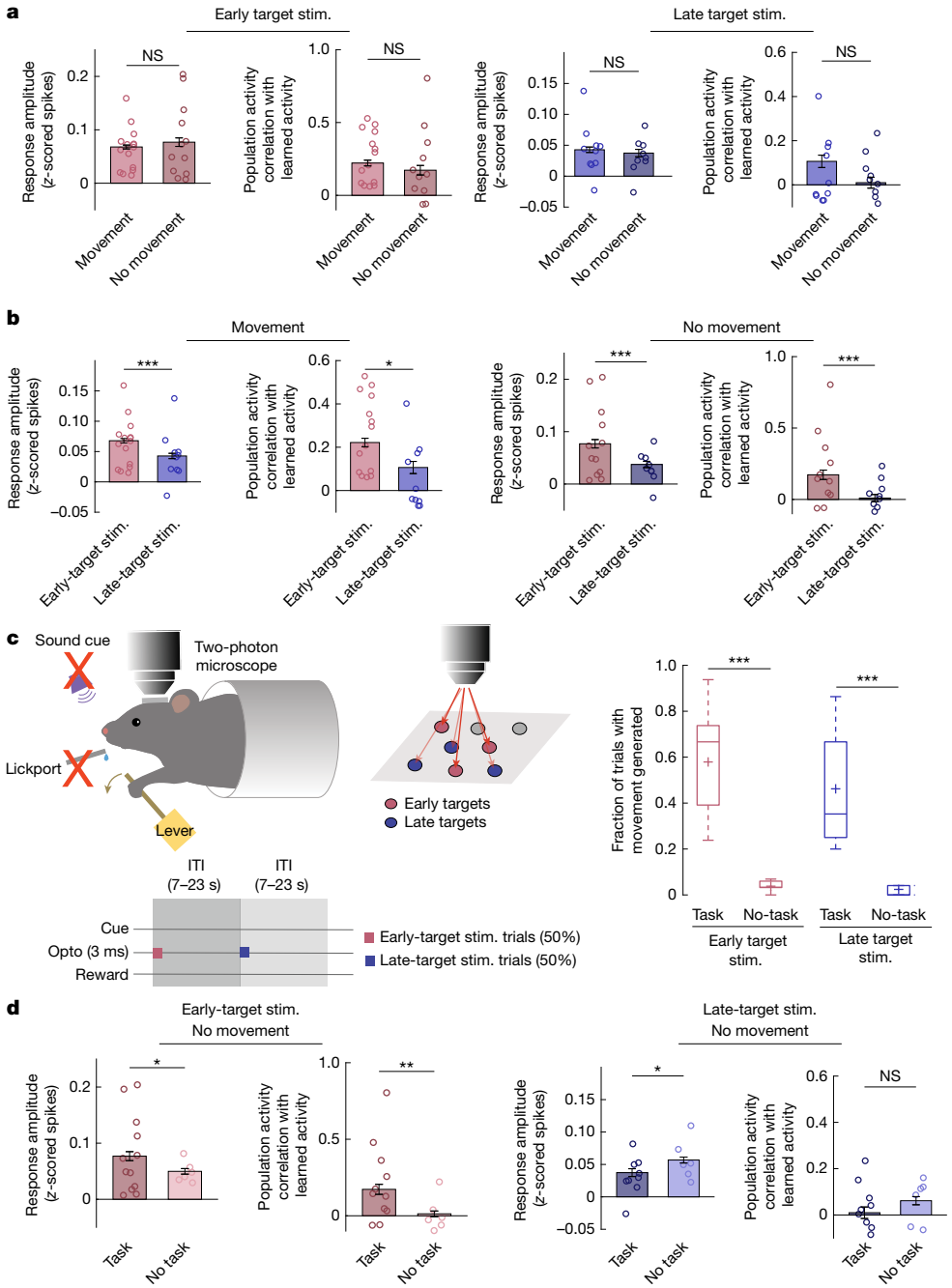


Fig. 5 | Early-target stimulation can induce learned activity without inducing movements in a context-dependent manner. a, Left, activity amplitude and correlation with learned activity of non-target neurons during movement trials compared with no-movement trials following early-target stimulation (amplitude: $P = 0.274$, correlation: $P = 0.251$ for trial type, mixed-effects model, two-sided). Activity of neurons was averaged during the 0.3 s following stimulation onset, averaged across trials. Right, same as left for late-target stimulation (amplitude: $P = 0.464$, correlation: $P = 0.800$ for trial type, mixed-effects model, two-sided). **b**, Left, activity amplitude and correlation with learned activity of non-target neurons during movement trials following early- and late-target stimulation (amplitude: $P = 1.27 \times 10^{-5}$, correlation: $P = 0.031$ for trial type, mixed-effects model, two-sided). Right, same as left for no-movement trials (amplitude: $P = 3.05 \times 10^{-4}$, correlation: $P = 2.92 \times 10^{-4}$ for trial type, mixed-effects model, two-sided). Same data as Fig. 5a, rearranged to highlight comparisons between early- and late-target stimulations. **c**, Left, schematic of no-task condition setup and task structure. Right, fraction of trials with movements generated within 0.3 s from trial onset (early task compared with early no-task: $P = 1.48 \times 10^{-11}$, late task compared with late no-task: $P = 6.68 \times 10^{-6}$ for trial type, mixed-effects model, two-sided). Early task: $n = 328/15/11$; early no-task: $n = 201/6/4$; late task: $n = 237/10/8$; late no-task: $n = 196/6/4$ (trials/sessions/mice). **d**, Left, activity amplitude and correlation with learned activity of non-target neurons during no-movement trials following early-target stimulation under task compared with no-task condition (amplitude: $P = 0.045$, correlation: $P = 0.008$ for trial type, mixed-effects model, two-sided). Right, same as left for late-target stimulation (amplitude: $P = 0.027$, correlation: $P = 0.112$ for trial type, mixed-effects model, two-sided). For this figure, bars and error bars show mean ± SEM; each dot represents an individual session mean. For response amplitude in **a**, **b**, **d**, early task movement: $n = 8{,}423/15/11$; early task no-movement: $n = 6{,}765/12/9$; early no-task no-movement: $n = 2478/6/4$; late task movement: $n = 5{,}120/10/8$; late task no-movement: $n = 4{,}502/8/7$; late no-task no-movement: $n = 2{,}478/6/4$ (neurons/sessions/mice). For correlation in **a**, **b**, **d**, early task movement: $n = 188/15/11$; early task no-movement: $n = 45/12/9$; early no-task no-movement: $n = 177/6/4$; late task movement: $n = 107/10/8$; late task no-movement: $n = 64/8/7$; late no-task no-movement: $n = 172/6/4$ (trials/sessions/mice).

manner, in which task engagement primes the network to selectively express the learned activity pattern in response to inputs to the early-onset neurons.

Discussion

In this study, we leveraged holographic stimulation to gain insights into the mechanisms that regulate the learned activity in M1 L2/3 and its role in generating learned movements. We found that brief and synchronous stimulation of approximately 20 early-onset, but not late-onset, movement-related neurons often triggered the learned activity within M1 L2/3 and induced the learned movement in a task-dependent manner. The efficacy of this stimulation depended on the initial state of the M1 L2/3 population activity. Taken together, we propose that the execution of the learned movement is ensured by two conditions: first, the motor cortex prepares the movement by entering the appropriate initial state of the population activity. Second, M1 receives specific inputs from other brain areas that activate the early-onset neurons. When both these two conditions are met, M1 generates the learned activity pattern that reliably drives the learned movement.

The importance of the initial state aligns well with the dynamical systems perspective on motor cortex movement control[28,30,31]. Previous studies have found that the pre-movement activity state of the motor cortex correlates with movement parameters[26,29,30], leading to the postulate that the motor cortex functions as a dynamical machine in which its state at one point in time predicts the next state. Our results provide direct support for this idea by demonstrating that the network response to our identical stimulus varies depending on the initial state of the network. However, the initial state does not completely specify the subsequent activity, as the external inputs also affect the evolution of network activity, as shown by the differences observed in our early-target and late-target stimulation. Future studies could explore whether targeted perturbations can correct suboptimal initial states to enhance movement consistency.

M1 exhibits dynamic activity patterns during movements. Whether M1 plays an active part in generating the activity pattern or passively reflects input activity has been debated[28,32–34]. By directly stimulating a subset of task-relevant neurons in M1 L2/3, we demonstrated that locally initiated cortical activity can 'pattern-complete' to drive the learned activity pattern, even in the absence of overt movements. We argue that these observations exclude the possibility that M1 is a purely passive machine that only reflects the temporally dynamic activity pattern provided by inputs from other brain areas. One possibility is that learning shapes the recurrent connectivity within M1 L2/3, forming a local circuit that can autonomously generate the learned activity pattern. In this possibility, activity may initiate in a group of neurons and then propagate through the network like a chain reaction—reminiscent of models proposed in zebra finch song production[35,36]. Furthermore, the inputs from the motor thalamus probably provide the triggering signal, activating the early-onset 'starter cells', and this thalamocortical connectivity refines during learning[36–41]. Computational studies have shown that recurrent neural networks can generate complex and initial-condition-dependent activity patterns that underlie movements[42–45]. M1 L2/3, as the primary input layer of M1[46], is highly recurrent in its connectivity and shows a high degree of learning-related plasticity[1,9–11,20–23], further supporting this possibility. Future research should explain the relative contributions of local and global interactions in cortical pattern generation and how particular circuits can be selectively primed in a task-dependent manner.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-11174-3.

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Methods

Animals

All procedures were conducted in compliance with protocols approved by the UCSD Institutional Animal Care and Use Committee and guidelines of the National Institutes of Health (NIH). C57BL/6 mice were acquired from Charles River Laboratory. Animals were group-housed in disposable plastic cages with standard bedding in a temperature-controlled room (about 21 °C) with a reversed light cycle. Experiments were performed during the dark period (10:00–22:00 hours). Males and females were randomly used for experiments. The experiments were done on young adults (8–15 weeks old).

Surgery and virus injections

Mice were anaesthetized with 3% isoflurane for induction, maintained at 1% throughout the surgical procedure. To prevent infection and inflammation, Baytril (10 mg kg^{-1}) and dexamethasone (2 mg kg^{-1}) were administered subcutaneously. A custom-built headplate was secured to the exposed skull using superglue and dental cement. A craniotomy (approximately 3 mm in diameter) was performed over the right caudal forelimb area of M1, centred at 0.3 mm anterior and 1.5 mm lateral from the bregma. Viral vectors (AAV8-hSyn-GCaMP6m-p2A-ChRmine-Kv2.1-WPRE, Addgene) were diluted to 1.6×10^{12} vg ml^{-1} and injected through a glass pipette at a depth of about 250 µm from the pia. Several injections (3–4) of 20–30 nl were made. Following injections, a glass window was implanted into the craniotomy and affixed with Vetbond (3 M), superglue and dental cement. At the end of the surgery, buprenorphine (0.1 mg kg^{-1}) was administered subcutaneously for postoperative pain management.

Behaviour

Water restriction began about 4 weeks post-surgery and about 15 days before behavioural training. The water volume was gradually reduced until stabilizing at 1 ml per day. Animal weights were monitored daily to ensure maintenance of at least 75% of their pre-restriction body weight.

Mice were trained to perform a lever-press task under head fixation one session per day. The lever consisted of a rotary encoder (E6B2-CWZ3E, YUMO) attached to a brass rod, which was held in place by two opposing springs and could be moved horizontally. The lever position was continuously recorded at 50,000 Hz using WaveSurfer and at 100 Hz using Bpod (Sanworks), integrated with custom code to monitor threshold crossings. Bpod also controlled the behavioural program. The rotary encoder has a native resolution of 1,024 units per 360° rotation, which was used for real-time threshold detection during behaviour within Bpod and offline analyses, and was subsequently converted to mm for lever position and mm s^{-1} for lever speed for data presentation and reporting (about 0.083 mm per encoder unit). Mice grasped the lever with their left paw while resting their right paw on a stationary thick rod, with their bodies in a plexiglass tube. A tone (2 kHz) signalled the cue period (up to 10 s), during which a successful lever press was rewarded with water (about 5 µl per trial). Each trial was followed by an inter-trial interval (ITI) of variable duration (7–13 s). A successful lever press was defined as the lever crossing a set threshold (about 0.83–1.25 mm) and remaining beyond the threshold for at least 1 s. Failure to achieve a rewarded movement during the cue period was not punished. Lever presses during ITI were neither rewarded nor punished. Each training session consisted of about 100 trials.

During the no-task condition (Fig. 5c), mice were first trained with the task for 2 weeks. These mice were then placed in the behavioural setup with the lever accessible, but the lickport was removed, and no sound cue or reward was provided. Optogenetic stimulation was applied to early targets and late targets in an interleaved manner, with each stimulation followed by a variable ITI (7–23 s).

In vivo two-photon imaging

Imaging was conducted with a commercial two-photon microscope (Ultima 2Pplus, Bruker) equipped with a 16×/0.8 NA objective (Nikon) and a Ti-Sa laser (MaiTai, Newport) tuned to 925 nm. Image acquisition was controlled by Prairie View Software (Bruker). Images (512 × 512 pixels at 1.5× magnification) were recorded at about 10 Hz per plane across three planes, spaced 30 µm apart in depth. The microscope was equipped with an electrotunable lens (Bruker) to enable imaging at different *z*-planes. Frame times were recorded and synchronized with behavioural recordings using the WaveSurfer software.

Holographic stimulation with a spatial light modulator during behaviour

A pre-opto session was conducted while mice performed the lever-press task under two-photon imaging. The pre-opto session included approximately half the number of trials in a regular training session to prevent the animals from reaching satiety before the opto session. Mice were then returned to their home cage while the neuronal activity was analysed offline to assess functional properties and determine which neurons to target in the subsequent opto sessions. After the analysis, mice were put back under the microscope, the same imaging fields were identified, and target neurons were manually selected for stimulation. Typically, the interval between the pre-opto and opto sessions was less than an hour. For early-target stimulation sessions, we sorted movement-active neurons by activity onset timing and selected from the earliest neurons that were visible and healthy. Similarly, for late-target stimulation sessions, we prioritized the neurons with later activity onsets. The activity onsets of target neurons were −9 ± 13 ms and 238 ± 24 ms (mean ± SEM) around movement onset in early-target and late-target stimulation sessions, respectively. We selected about 20 neurons for stimulation based on the observation that stimulating fewer than 20 neurons rarely elicited movements (Fig. 1f). Moreover, in most experiments, we did not have many more than 20 early-onset neurons that we were able to visually identify quickly while defining the spatial patterns for stimulation.

In opto sessions, the trial order was determined using a pseudorandom sequence to ensure all trial types were distributed throughout the session rather than clustered in specific periods. Specifically, we generated a sequence of 30 trials containing a randomized mixture of the three trial types (voluntary, stimulation and omission), and this sequence was repeated to form the full session (typically about 100 trials).

For holographic stimulation, the microscope and acquisition system (Bruker) was used with the NeuraLight 3D Spatial Light Modulator (SLM) (Bruker) and a second, 1,035-nm high power laser (Monaco 1035-40-40, Coherent). Spatial accuracy of SLM targeting was ensured through built-in calibration routines before each stimulation session. The stimulation laser operated at a pulse repetition rate of 1 MHz and spirally scanned over each target cell body (10 µm diameter, 5 revolutions), delivering about 7 mW of power per cell for 3 ms in each opto trial. To further quantify the degree of stimulation spatial accuracy (Extended Data Fig. 1e), we stimulated a fixed set of five target cells while systematically shifting the stimulation centre relative to the target location by up to 60 µm. Activity of target neurons was averaged over the first 0.5 s following stimulation onset.

Data analysis

Behaviour analysis. Lever position was recorded by both Bpod and WaveSurfer. WaveSurfer recordings were used whenever available. Lever position was sampled at 100 Hz. For movement analysis, we excluded trials that started during ongoing movements, defined as lever speeds exceeding 12.5 mm s^{-1} during the 50 ms before trial onset. Based on this criterion, 17.4 ± 1.7% (mean ± SEM) of trials were excluded. For trials with no ongoing movements, movement onset was defined

as the first time point after trial onset at which lever velocity exceeded 41.7 mm s^{-1} in the forward direction. This threshold was empirically defined to capture meaningful movements (Extended Data Fig. 2a) and corresponds to five encoder units within a 10 ms window. Reaction time was calculated as the interval from trial onset to movement onset. Only the trials with a movement onset within 0.3 s of trial onset were considered movement trials. We imposed this constraint because movements occurring much later than the stimulation in opto trials were unlikely to be triggered by the stimulation. The specific 0.3 s time window was chosen based on the mean reaction time of 0.24 s observed in expert mice during voluntary movements (Extended Data Fig. 1b). For consistency, the same criterion was applied to movements in all trial types.

For analyses of no-movement trials (Fig. 5), we applied stricter criteria to ensure the absence of overt movements. The threshold for movement onset was defined as the first time point after trial onset at which the lever velocity exceeded 25.0 mm s^{-1} in the forward direction, corresponding to a change of <3 encoder units per 10 ms window. Trials in which no such movement onset occurred within 1 s following stimulation were included in these analyses as no-movement trials.

Movement analysis was done on lever time series for 1 s starting from 0.1 s before movement onset. The learned movement pattern was defined for each session separately to account for the idiosyncrasies of individual animal behaviour. We first identified highly stereotyped movements as those with a correlation above 0.7 with the mean of all voluntary movements. A random half of these stereotyped movements were averaged to form the learned movement. For comparisons involving the learned movement, the trials used in their construction were excluded from the analysis. Movement similarity was measured by the correlation coefficient of the lever displacement over the 1-s period, quantifying the temporal pattern of movements. As a result, variability in the hold period, such as failures to sustain the hold, can reduce measured correlation with the learned movement.

Image analysis

Regions of interest identification and fluorescence analysis. Suite2P software[47] was used for motion correction and extraction of regions of interest (ROIs), which correspond to individual neurons. ROI identification was further refined by manual selection. Fluorescence signals from ROIs were deconvolved to remove fluorescence decay and infer spiking activity using the Suite2P built-in deconvolution algorithm. The estimated spike activity was used for all neural activity analyses.

To match neurons between pre-opto sessions and opto sessions, ROIMatchPub (https://github.com/ransona/ROIMatchPub) was used for automatic detection, followed by manual matching of target neurons with custom code.

Movement-related neuron classification. Movement-related neurons were classified as previously described[1,10,12]. Briefly, the dot product of the binarized lever movement time series (movement epochs compared with non-movement epochs) and estimated spikes was calculated for each neuron. This value from actual data was then compared with the dot products when shuffling the time points of movement epochs 1,000 times. Actual values above the 95th percentile of the shuffled distribution were classified as 'movement-active', whereas those below the 5th percentile were classified as 'movement-suppressed'. All other neurons were considered as 'non-movement-related'.

Movement-related activity onset. To estimate the activity onset timing of individual neurons during movements, we aligned the activity of each neuron to movement onsets. We considered only movements that did not have another movement within 1 s before movement onset to avoid contamination of activity related to the other movements. Activity averaged across movements was then used for onset detection. Activity onset was defined as the first frame between −0.2 s before movement onset to end of movement in which activity was higher than baseline activity (average activity between −1 s and −0.5 s before movement onset) by more than one standard deviation for 200 ms.

Non-target neuron population activity pattern. Activity of non-target neurons was averaged during the 1-s period following cue or stimulation onset. Time-averaged activity of each neuron was then averaged across learned movements, opto-induced movements or voluntary movements (Figs. 1h,i, and 2f and Extended Data Fig. 7b,c).

Non-target neuron recruitment. Response amplitudes of non-target neurons to optogenetic stimulation of target neurons were calculated for each non-target neuron in each trial as the difference between the average activity during the 1-s period following stimulation onset and the average activity during the 1-s period preceding stimulation onset. Non-target neurons with response amplitudes exceeding 0.2 *z*-scored spikes were classified as responsive to, and thus recruited by, the optogenetic stimulation of the target neurons (Fig. 2e and Extended Data Fig. 7d).

Target neuron general properties. The anatomical distribution of targets was quantified for each session as the fraction of targets along the imaging depth, anterior–posterior and medial–lateral axes, and these values were subsequently averaged across sessions. The degree of spatial clustering was quantified by computing, for each target neuron, the distance to its nearest-neighbour target neuron. The resulting nearest-neighbour distance distribution was then compared with the chance distribution generated by shuffling target labels 1,000 times. Peak amplitudes of movement-related activity were quantified as the maximum neural activity within the −1 s to 2 s window around movement onset. Response consistency was defined as the fraction of movements in which the activity of the given neuron within the 2-s window following movement onset was significantly higher than its baseline activity. For both peak amplitudes and response consistency measurements, we used movements recorded during the pre-opto session in the absence of any stimulation (Extended Data Fig. 6).

Principal component analysis. For each session, we compiled a data matrix for voluntary trials, X, of size $n \times \mathrm{fr} \times \mathrm{tr}$, where n is the number of non-target neurons, fr is the 40 image frames corresponding to −2 s to 2 s around cue onset, and tr is the total number of voluntary trials. PCA was applied to trial-averaged data in X (size $n \times \mathrm{fr}$), and the principal components (PCs) explaining 80% of the variance in neural space were extracted. We then projected population activity of individual trials to the PCs explaining 80% of total variance (Figs. 3–5 and Extended Data Fig. 8). In Extended Data Fig. 8, n is the number of all neurons. In Extended Data Fig. 9a–c, tr is half the number of learned movement trials.

Movement–activity relationship. To calculate the trajectory distance, we computed the Euclidean distance in the PC space between trajectories at each time point and these values were added across time points. For pre-cue/pre-stimulation activity, we averaged the distance over the 500 ms preceding trial onset, whereas for post-cue/post-stimulation activity, we averaged the distance during the 1 s following trial onset to obtain the trajectory distance for each trial. To construct the learned trajectory, we averaged the projected activity from the trials that contributed to the learned movement. The activity from these trials was then excluded from further analyses to avoid self-comparison (Figs. 3 and 4 and Extended Data Figs. 8 and 9).

No-movement-trial analysis. Response amplitudes were calculated as the trial-averaged activity of non-target neurons averaged during the 0.3-s period following stimulation onset. To calculate correlation with learned activity, we computed the Pearson correlation between the projected population activity averaged over the 0.3-s window following stimulation onset in opto trials and that of the learned movement (Fig. 5).

Movement decoding analysis

To examine the relationship between movement-related activity and movements, we sought to decode across-trial variability of lever movements using across-trial variability of population activity.

We aligned the temporal resolution of lever traces with that of the neural activity by downsampling lever traces from 100 Hz to 10 Hz by averaging every 10 data points. We divided voluntary trials into 80% for training the model and 20% for testing. Opto trials were not included in training.

To define variability in lever movements, we used the difference between the lever movement on each trial and the learned movement. Similarly, for population activity, we used the difference of the trial activity from the learned activity. These differences from the learned patterns, or 'residuals', were used instead of the original activity and movement to avoid building a model that only predicts the trial-averaged movement and does not explain across-trial variability.

The population activity residuals were first processed by PCA, concatenating activity across trials to capture trial-by-trial variability. The top PCs capturing 80% of the total variance were selected. Using these PC activities in the training set, we built a LASSO regression model to predict lever residuals. The optimal λ value was determined through 10-fold cross-validation. The trained model was tested on the test voluntary trials and opto trials. Model performance was evaluated by calculating correlation coefficients between actual and predicted lever residuals on a trial-by-trial basis.

To test statistical significance, a control regression model was built. Lever trace residuals were randomly shuffled across trials while preserving their temporal structure, whereas neural residuals retained their original trial order. The lever trace residuals were shuffled such that trials were never retained in their original positions. The control model was then trained and tested as above.

Finally, we compared the correlation coefficients from the main model with those obtained from the control model using a mixed-effects model, accounting for session effects. Moreover, model performance for test voluntary trials and opto trials was evaluated within the same mixed-effects framework.

Movement probability analysis

To test whether initial state predicts movement probability, we trained a linear decoder on voluntary trials and tested its performance in opto and omission trials (Extended Data Fig. 9d).

Trials were classified as movement or no-movement using the same criteria as Fig. 5. For each trial, we extracted population activity during the 0.5 s preceding cue onset (voluntary trials), stimulation onset (opto trials) or trial onset (omission trials), and averaged the activity of each neuron across this window to yield one pre-trial activity value per neuron per trial. Sessions with fewer than two voluntary trials in either movement class were excluded from further analysis.

For each session, we trained a lasso-regularized logistic regression decoder to predict trial-by-trial movement from pre-trial population activity in voluntary trials. Features were first projected onto the top three principal components of the training data to reduce dimensionality; the PCA basis was fit exclusively on the training set and then applied to held-out trials to prevent leakage. The λ value was selected by 5-fold cross-validation. The resulting decoder weights were then frozen and applied to opto and omission trials to compute a linear predictor reflecting a movement-ready initial state.

We fit separate mixed-effects logistic regression models relating this linear predictor to the actual movement outcome of each trial, with random intercepts for session nested within mouse. The relationship between initial state and movement outcome was assessed from the slope coefficient of the model for each condition. To test whether the opto and omission relationships differed from one another, we pooled trials from both trial types and used a likelihood ratio test to compare a mixed-effects model allowing both slope and offset to vary by trial type against a reduced model with no trial type effect.

Statistics

All statistical analyses were performed in MATLAB. Statistical tests used are described in the figure legends. Statistical significance was determined by a standard threshold of *P*-values less than 0.05. Results are denoted in the figures as follows: $*P < 0.05$, $**P < 0.01$ and $***P < 0.001$.

For analyses pooling trials across sessions, we used mixed-effects models (using the fitlme function in MATLAB) to account for the nested data structure and minimize the effects of session-to-session variability. The models used in this study are as follows:

$$y \sim \text{trial type} + (1|\text{mouse} : \text{session})$$

where the fixed effect is the trial type, voluntary or early-target stimulation or late-target stimulation or omission trials, movement or no-movement trials, task trials or no-task trials, and the random effect is experimental session nested within each mouse. This model was used to compare the average lever speed and total lever displacement (Extended Data Fig. 4a), fraction of trials with movement generated (Figs. 1f, 2b and 5c), correlation with learned movement (Fig. 2c and Extended Data Fig. 4b), reaction time (Fig. 2d and Extended Data Fig. 4b), fraction of responsive non-target neurons (Fig. 2e and Extended Data Figs. 5a and 7d), response amplitude of non-target neurons (Fig. 2e and Extended Data Figs. 5a and 7d), population activity amplitudes (Fig. 5a,b,d), population activity correlation with learned activity (Fig. 5a,b,d), activity peak amplitudes and response consistency during movement (Extended Data Fig. 6c), between pairs of trial types.

$$y \sim \text{learned activity} + (1|\text{mouse} : \text{session})$$

where the fixed effect is the learned activity, and the random effect is the experimental session nested within each mouse. This model was used to assess the relationship between learned activity and opto-induced or voluntary movement activity (Figs. 1i and 2f and Extended Data Fig. 7c).

$$y \sim \text{distance} + (1|\text{mouse} : \text{session})$$

where the fixed effect is the population activity distance, and the random effect is the experimental session nested within each mouse. This model was used to assess relationship between population activity distance to learned activity and correlation with learned movement (Figs. 3a,b and 4a,b and Extended Data Figs. 8 and 9a), relationship between trial-to-trial population activity distance and trial-to-trial movement correlation (Figs. 3c and 4d and Extended Data Fig. 9c), and the relationship between pre-stimulation/pre-cue population activity distance to learned activity and post-stimulation/post-cue population activity distance to learned activity, respectively (Fig. 4c and Extended Data Fig. 9b).

$$y \sim \text{condition} + (1|\text{mouse} : \text{session})$$

where the fixed effect is the condition, voluntary or shuffled (voluntary) or opto or shuffled (opto), and the random effect is the experimental session nested within each mouse. This model was used to compare the correlation coefficients between actual and predicted lever residuals from different regression models (Fig. 3f).

$$y \sim \text{group} \times \text{location} + (1|\text{mouse} : \text{session})$$

where the fixed effects were group, location and their interaction (group × location), and the random effect was the experimental session

nested within each mouse. The interaction term was used to evaluate whether the two groups exhibited comparable spatial distributions of target neurons across cortical depth (Extended Data Fig. 6a, left), anterior–posterior position (Extended Data Fig. 6a, middle) and medial–lateral position (Extended Data Fig. 6a, right).

$$y \sim \text{data type} + (1|\text{mouse} : \text{session})$$

where the fixed effect was data type and the random effect was the experimental session nested within each mouse. This model was used to evaluate differences in spatial clustering between early-target neurons and shuffled data (Extended Data Fig. 6b, left), between late-target neurons and shuffled data (Extended Data Fig. 6b, middle) and between early- and late-target neurons (Extended Data Fig. 6b, right).

$$y \sim \text{initial state} + (1|\text{mouse} : \text{session})$$

where the fixed effect was the movement-ready initial state obtained from decoder output and the random effect was the experimental session nested within each mouse. The fitglme function in MATLAB was used. This model was used to evaluate the relationship between initial state and movement probability (Extended Data Fig. 9d).

$$y \sim \text{initial state} \times \text{trial type} + (1|\text{mouse} : \text{session})$$

where the fixed effects were initial state, trial type (opto or oms), and their interaction (initial state × trial type), and the random effect was the experimental sessions nested within each mouse. The fitglme function in MATLAB was used. This model was compared against a reduced model excluding trial type effects to assess whether the separate logistic fits for opto and oms trials differed significantly (Extended Data Fig. 9d).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data used in this study are available at Zenodo (https://doi.org/10.5281/zenodo.22216381)[48]. Source data are provided with this paper.

Code availability

The code used in this study is available at Zenodo (https://doi.org/10.5281/zenodo.22216381)[48].

47. Pachitariu, M. et al. Suite2p: beyond 10,000 neurons with standard two-photon microscopy. Preprint at *bioRxiv* https://doi.org/10.1101/061507 (2017).
48. Chen, Q. et al. Targeted stimulation of motor cortex ensembles drives learned movements. *Zenodo* https://doi.org/10.5281/zenodo.22216381 (2026).

Acknowledgements We dedicate this manuscript to our friend and colleague, A.W., and acknowledge her surviving parents F. Wu and F. Qu. We thank B. Morales, A. Medina, D. Arakelyan and E. Hall for technical assistance and other Komiyama Lab members, especially J. Li and E. Gjoni, for their discussion. We also thank all our friends who have reached out to us to offer support during the difficult time.

Author contributions A.W. and T.K. conceptualized the study. A.W., Q.C., T.K. and J.A. devised the methodology. Q.C., A.W., B.Y., Z.T. and A.R. conducted the investigation. Q.C. helped with visualization. Q.C., A.W., and S.C. performed the formal analysis. Q.C. validated the study. A.W., Q.C. and T.K. helped with the resources. Q.C. and A.W. handled the software. Q.C. curated the data. T.K. helped with funding acquisition. T.K. was in charge of project administration. T.K. supervised the study. Q.C. and T.K. wrote the original draft. Q.C., T.K., J.A., B.Y., S.C., Z.T. and A.R. reviewed and edited the paper.

Funding This work was supported by the following funding sources: the NIH grants R01 NS125298, R01 NS091010, R01 DC018545 and R01 MH128746 (T.K.); the National Science Foundation grant 2024776 (T.K.); and the Simons Collaboration on the Global Brain Pilot Award (T.K.).

Competing interests The authors declare no competing interests.

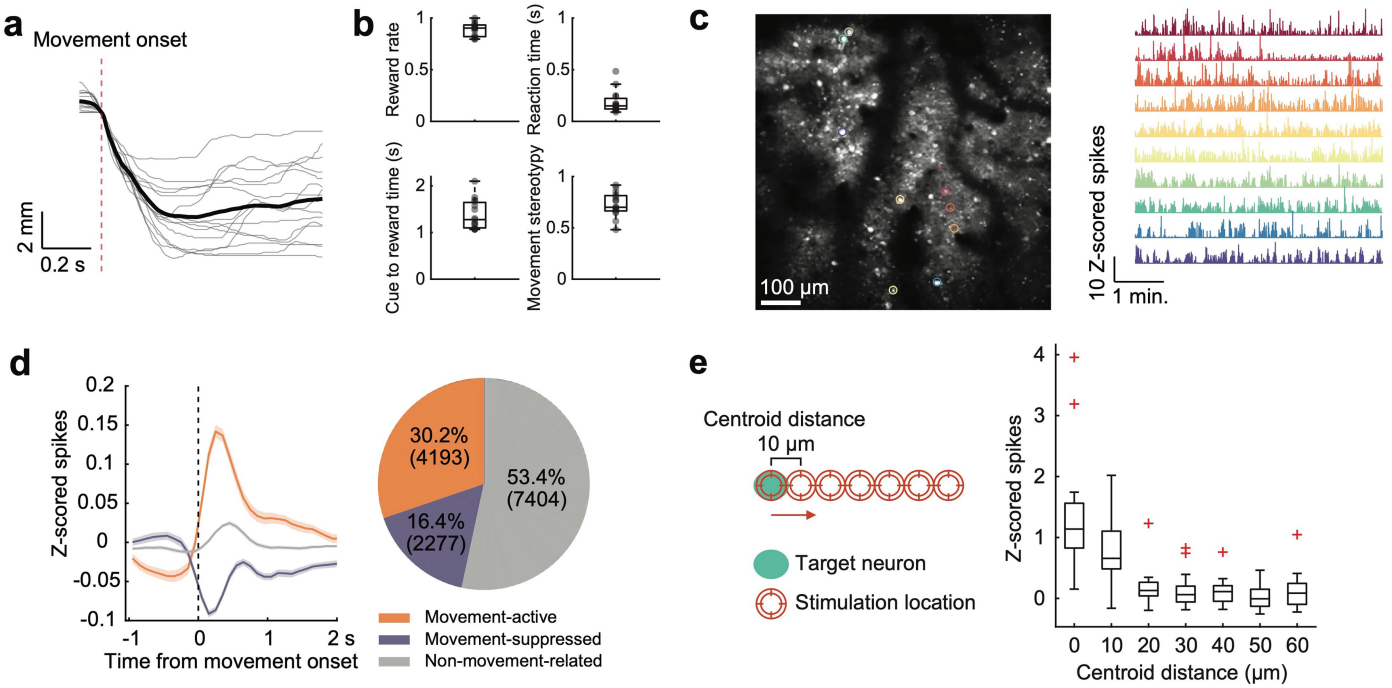
Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-026-11174-3.

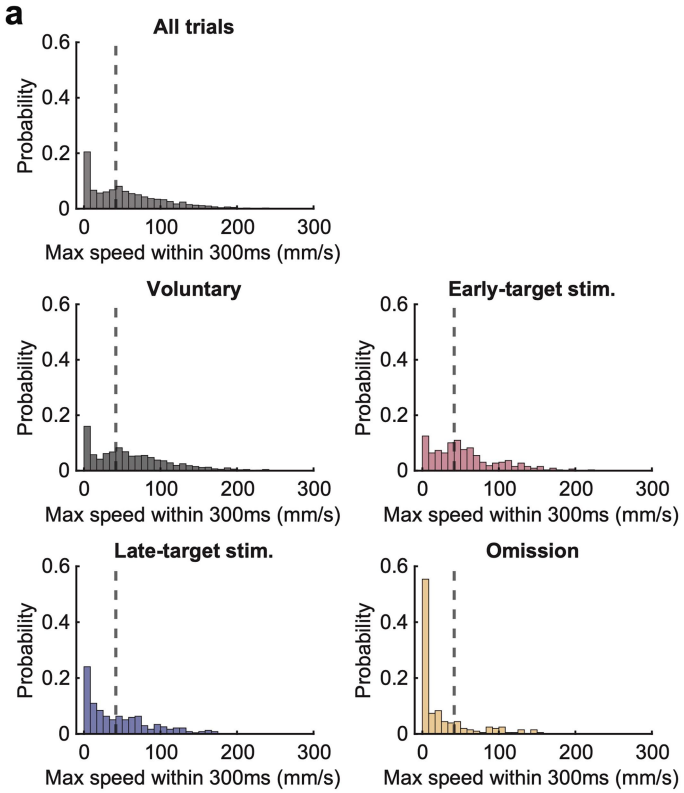
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Peer review information *Nature* thanks Kayvon Daie and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

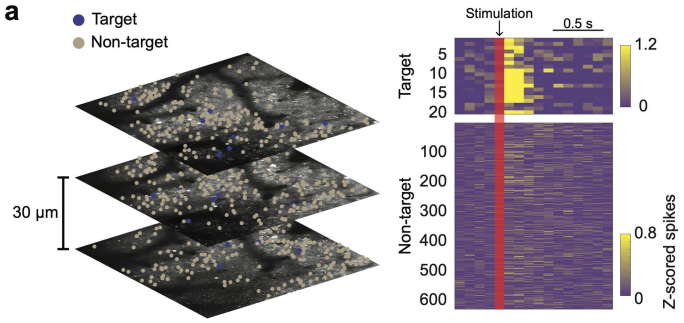
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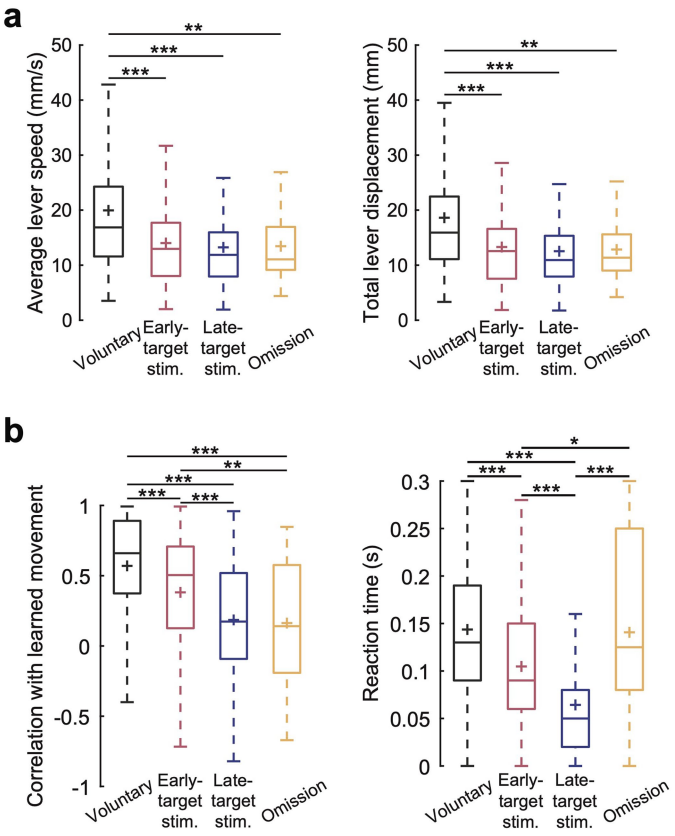
Extended Data Fig. 1 | Additional information on experimental setup. a. Example lever trajectories of rewarded trials in an expert session from one mouse. Grey, individual trials; black, mean. b. Reward rate, reaction time, cue onset to reward time, and movement stereotypy (median trial-to-trial correlation coefficients of lever movements for each session) at expert stage (N = 15 mice). Box and whisker plots: median and interquartile range. c. Left, example field of view of in vivo two-photon calcium imaging. Right, z-scored time series of estimated spikes from example neurons outlined in the image in the middle. This panel is representative of all 19 animals imaged in this study. d. Left, average population activity of all movement-active, movement-suppressed and non-movement-related neurons. Mean ± SEM across session-averages. Right, proportions of movement-active, movement-suppressed and non-movement-related neurons. N = 13874/25/15 (neurons/sessions/mice). e. Stimulation spatial accuracy. Left, schematic of experimental design. Stimulation locations were positioned at increasing distances from the target neuron while recording target neuron activity. Right, trial-averaged activity of target neurons at various centroid distances between target neuron and stimulation location. Box and whisker plots: median and interquartile range. N = 20/4/1 (targets/sessions/mice).



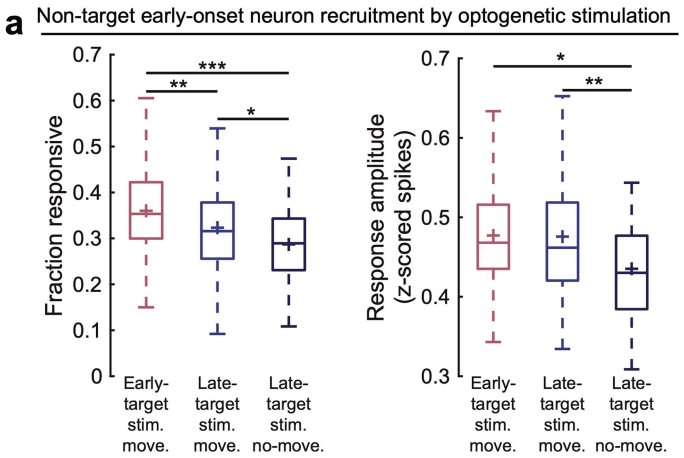
Extended Data Fig. 2 | Movement trial classification. a. Histograms of the maximum lever speed within 300 ms of trial onset for all trials, voluntary trials, early-target stimulation trials, late-target stimulation trials, and omission trials, respectively. Vertical dashed lines indicate the thresholds used to define movement trials. No-movement trials used in Fig. 5 were defined with a stricter criterion (Methods). Voluntary: N = 1199/25/15; early: N = 328/15/11; late: N = 237/10/8; omission: N = 204/25/15 (trials/sessions/mice).



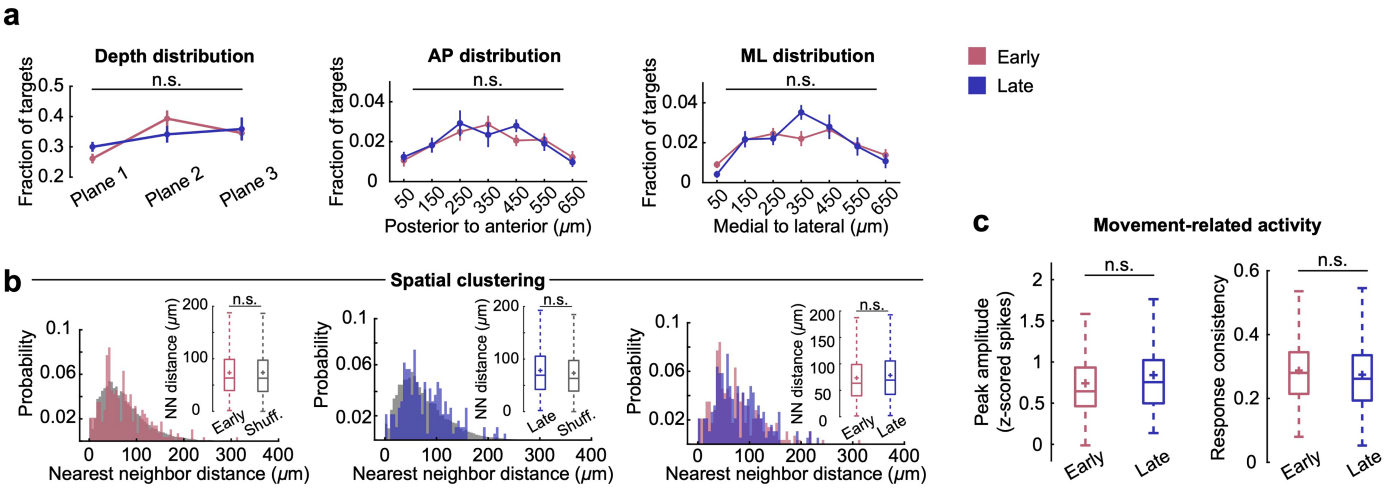
Extended Data Fig. 3 | Example late-target stimulation. a. Left, example two-photon calcium imaging of three planes in M1 L2/3 spanning 60 µm in depth, showing target neurons (blue) and non-target neurons (beige). Right, trial-averaged responses of late-onset target and non-target neurons in opto trials aligned to stimulation onset. Red vertical bar indicates stimulation timing.



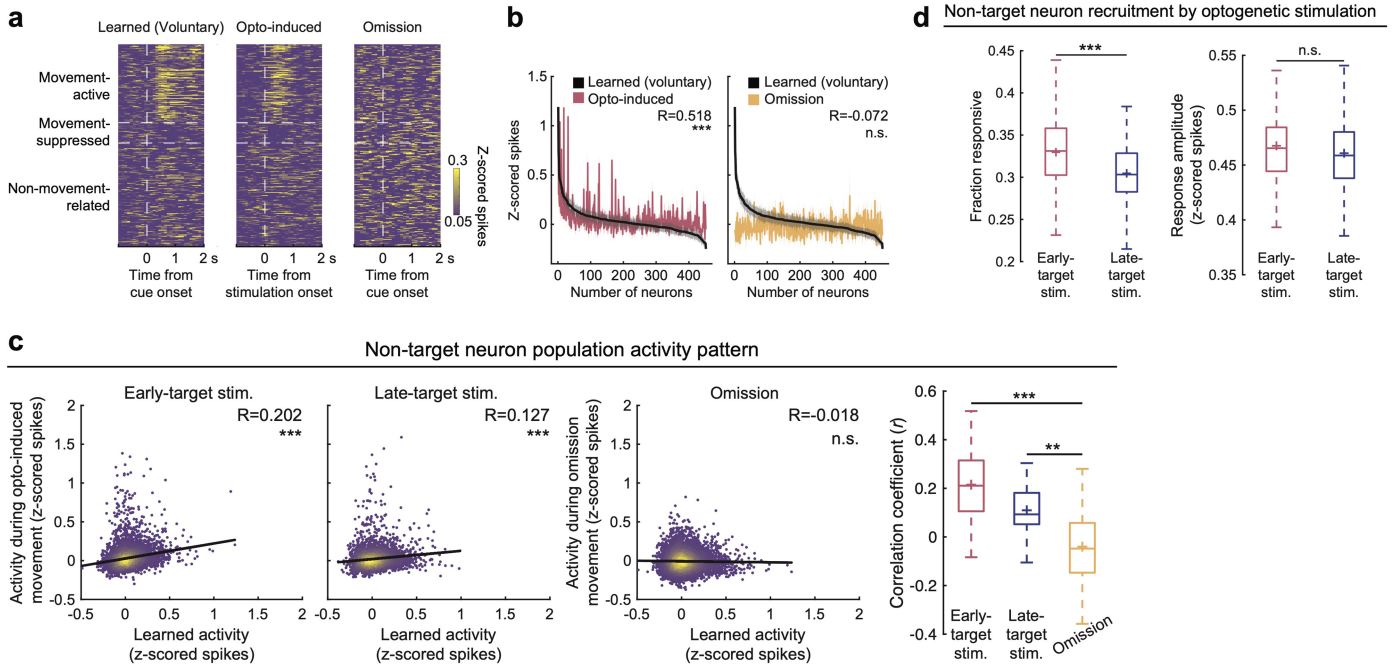
Extended Data Fig. 4 | Additional movement quantification. a. Left, average lever speed over the 0.9-sec period starting from cue/stimulation onset in movement trials in each trial type (voluntary vs early: p = 1.81 × 10-8, voluntary vs late: p = 1.64 × 10-14, voluntary vs. omission: p = 3.81 × 10-3, early vs late: p = 0.430, early vs. omission: p = 0.679, late vs. omission: p = 0.442 for trial type, mixed-effects model, two-sided). Right, total lever displacement in each trial type (voluntary vs early: p = 3.12 × 10-8, voluntary vs late: p = 1.66 × 10-14, voluntary vs. omission: p = 4.33 × 10-3, early vs late: p = 0.403, early vs. omission: p = 0.717, late vs. omission: p = 0.430 for trial type, mixed-effects model, two-sided). Voluntary: N = 729/25/15; early: N = 184/15/11; late: N = 107/10/8; omission: N = 42/25/15 (trials/sessions/mice). b. Left, correlation with the learned movement in each trial type (voluntary vs early: $p = 5.24 \times 10^{-12}$, voluntary vs late: $p = 1.23 \times 10^{-12}$, voluntary vs omission: $p = 7.41 \times 10^{-11}$, early vs late: $p = 2.45 \times 10^{-4}$, early vs omission: $p = 0.003$, late vs omission: $p = 0.900$ for trial type, mixed-effects model, two-sided). Right, reaction time of movements in each trial type (voluntary vs early: $p = 1.10 \times 10^{-11}$, voluntary vs late: $p = 1.55 \times 10^{-30}$, voluntary vs omission: $p = 0.571$, early vs late: $p = 1.98 \times 10^{-6}$, early vs omission: $p = 0.034$, late vs omission: $p = 2.94 \times 10^{-8}$ for trial type, mixed-effects model, two-sided). Voluntary: N = 489/25/25; early: N = 184/15/11; late: N = 107/10/8; omission: N = 42/25/15 (trials/sessions/mice). For this figure, box-and-whisker plots: median and interquartile range, + indicates mean. Same analysis as in Fig. 2c,d, except that omission trials are included for comparison.



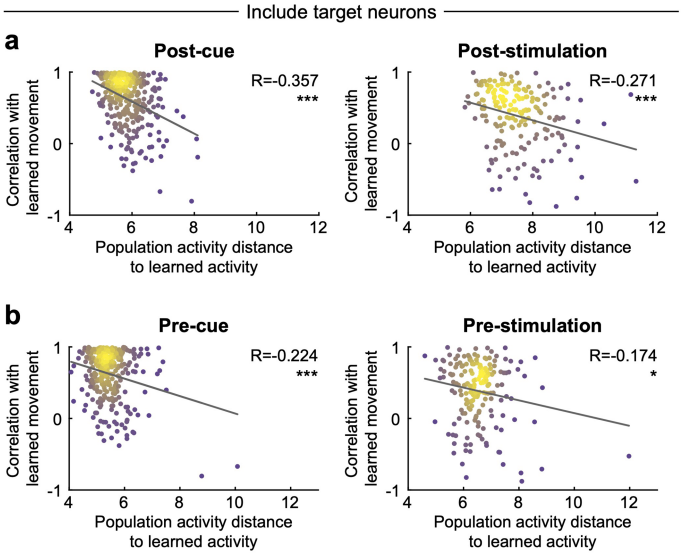
Extended Data Fig. 5 | Non-target early-onset neuron recruitment by optogenetic stimulation. a. Left, fraction of responsive non-target early-onset neurons per trial for early-, late-target stimulation movement trials and late-target stimulation no-movement trials (early move. vs. late move.: $p = 1.95 \times 10^{-3}$; early move. vs. late no-move.: $p = 5.40 \times 10^{-8}$; late move. vs. late no-move.: $p = 0.015$ for trial type, mixed-effects model, two-sided). Right, response amplitudes of the responsive non-target early-onset neurons for early-, late-target stimulation movement trials and late-stimulation no-movement trials (early move. vs. late move.: $p = 0.875$; early move. vs. late no-move.: $p = 0.013$; late move. vs. late no-move.: $p = 1.87 \times 10^{-3}$ for trial type, mixed-effects model, two-sided). Early move.: N = 188/15/11; late move.: N = 107/10/8; late no-move.: N = 64/10/8 (trials/sessions/mice). Box and whisker plots: median and interquartile range, + indicates mean. This figure uses the same analysis procedure as Fig. 2e, except that only non-target early-onset neurons are examined, rather than all non-target neurons.



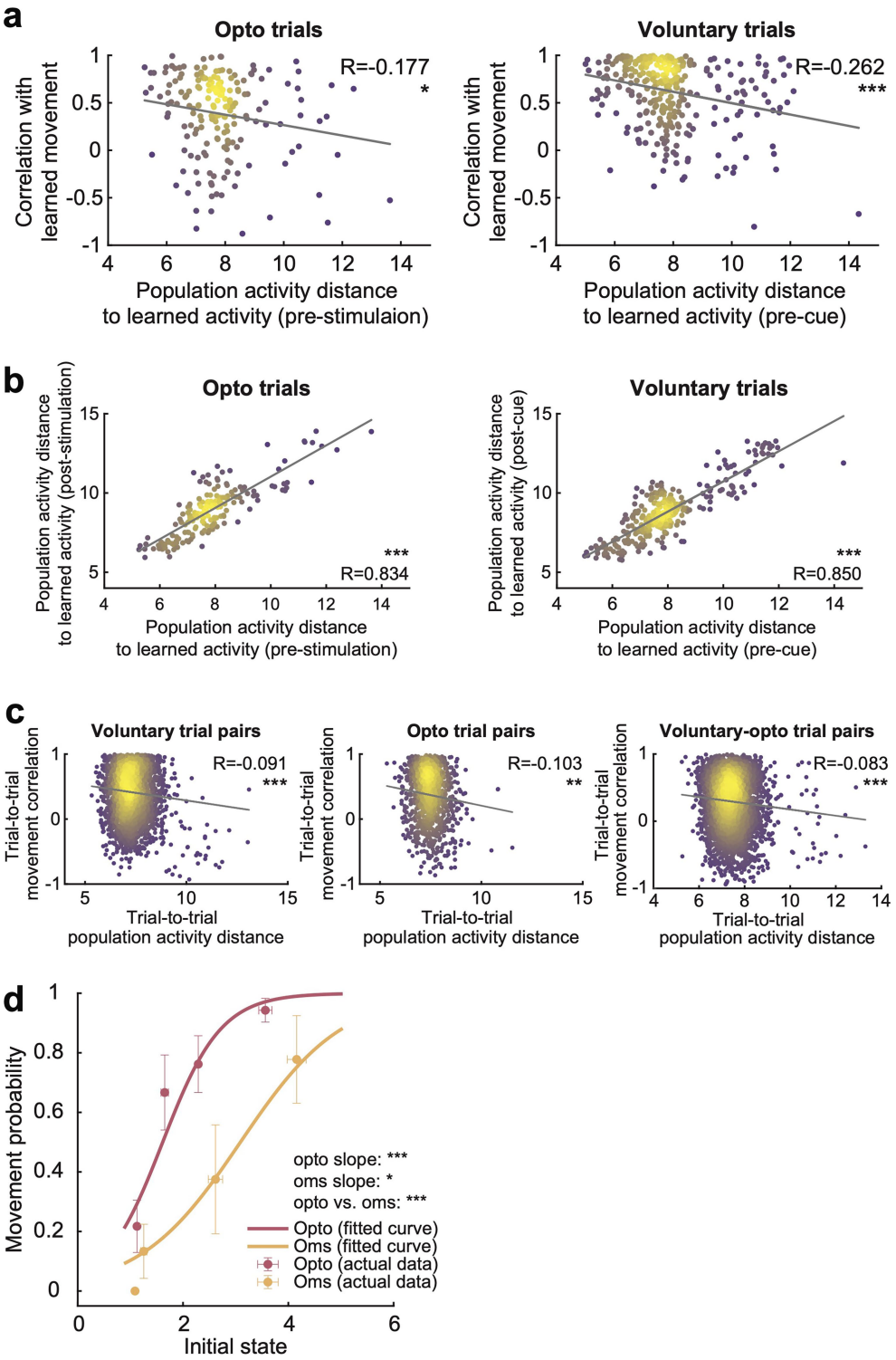
Extended Data Fig. 6 | General properties of target neurons do not explain their distinct stimulation outcomes. a. Left, fraction of early- or late-onset targets located in different imaging depths ($p = 0.153$ for group × location, mixed-effects model, two-sided). Middle, fraction of early- or late-onset targets located along the anterior-posterior axis within the field of view ($p = 0.667$ for group × location, mixed-effects model, two-sided). Right, fraction of early- or late-onset targets located along the medial-lateral axis ($p = 0.813$ for group × location, mixed-effects model, two-sided). Mean ± SEM. Early: N = 15/11; late: N = 10/8 (sessions/mice). b. Left, the distribution of nearest neighbor distances between early-onset targets and a shuffled control ($p = 0.930$ for data type, mixed-effects model, two-sided). Middle, same as left for late-onset ($p = 0.133$ for data type, mixed-effects model, two-sided). Right, Comparison between early- and late-onset targets ($p = 0.262$ for target type, mixed-effects model, two-sided). Early: N = 285/15/11; shuffled (early): N = 285000/15/11; late: N = 188/10/8; shuffled (late): N = 188000/10/8 (neurons/sessions/mice). c. Left, the peak activity amplitude of early- vs. late-onset targets during voluntary movements ($p = 0.351$ for target type, mixed-effects model, two-sided). Right, the response consistency of early- vs. late-onset targets during voluntary movements ($p = 0.516$ for target type, mixed-effects model, two-sided). Response consistency was measured as the fraction of movements during which a given neuron was active. Early: N = 285/15/11; late: N = 188/10/8 (neurons/sessions/mice). For this figure, box and whisker plots: median and interquartile range, + indicates mean.



Extended Data Fig. 7 | Population activity including no-movement trials, with direct comparison to omission trials. This figure is analogous to Figs. 1h,i and 2e,f, with the difference that all trials were included for each trial type, including no-movement trials. a. Activity of movement-active, movement-suppressed, and non-movement-related neurons (separated by horizontal dashed lines) averaged across learned movement trials, opto trials, and omission trials separately, from an example session. Each row represents a non-target neuron. b. Average activity of non-target neurons during the 1 sec following cue/stimulation onset sorted by amplitudes in learned movements from the same example session (opto-induced: $p = 2.26 \times 10^{-32}$, R = 0.518; omission: $p = 0.125$, R = −0.072. Pearson's correlation, two-sided). Mean ± SEM. c. Leftmost, average activity of non-target neurons during learned movements versus in early-target stimulation trials pooled across sessions ($p = 1.59 \times 10^{-84}$ for learned activity, mixed-effects model, two-sided; Pearson's correlation, R = 0.202). Each dot represents the activity of one neuron averaged during the 1 sec following cue/stimulation onset, averaged across trials. Center-left, average activity of non-target neurons during learned movements versus in late-target stimulation trials pooled across sessions ($p = 2.44 \times 10^{-20}$ for learned activity, mixed-effects model, two-sided; Pearson's correlation, R = 0.127). Center-right, average activity of non-target neurons during learned movements versus in omission trials pooled across sessions ($p = 0.075$ for learned activity, mixed-effects model, two-sided; Pearson's correlation, R = −0.018). Rightmost, correlation coefficients between non-target neuron population activity in learned movements and early-, late-target stimulation, and omission movements calculated for individual sessions (early vs. late: $p = 0.081$, early vs. omission: $p = 5.10 \times 10^{-5}$, late vs. omission: $p = 4.15 \times 10^{-3}$; Wilcoxon rank-sum test, two-sided). Early: N = 8789/15/11; late: N = 5252/10/8; omissions: N = 14041/25/15 (neurons/sessions/mice). d. Left, fraction of responsive non-target neurons per trial for early-target and late-target stimulation trials ($p = 1.85 \times 10^{-4}$ for trial type, mixed-effects model, two-sided). Right, response amplitudes of the responsive non-target neurons for early-target and late-target stimulation trials ($p = 0.160$ for trial type, mixed-effects model, two-sided). Early: N = 328/15/11; late: N = 237/10/8 (trials/sessions/mice). For this figure, box and whisker plots: median and interquartile range, + indicates mean.

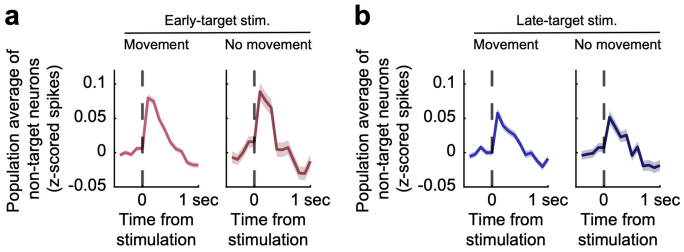


Extended Data Fig. 8 | Activity-movement relationship analysis including target neurons. These are equivalent to Figs. 3a,b and 4a,b except that the analysis includes target neuron activity. a. Left, post-cue population activity distance to the learned activity negatively correlated with correlation with the learned movement in voluntary trials ($p = 6.84 \times 10^{-11}$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.357). Each dot represents an individual trial. Right, post-stimulation population activity distance to the learned activity negatively correlated with correlation with the learned movement in opto trials ($p = 1.79 \times 10^{-4}$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.271). Post-cue: N = 313/15/11; post-stimulation: N = 184/15/11 (pairs/sessions/mice). b. Left, pre-cue population activity distance to the learned activity negatively correlated with correlation with the learned movement in voluntary trials ($p = 6.06 \times 10^{-5}$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.224). Each dot represents an individual trial. Right, pre-stimulation population activity distance to the learned activity negatively correlated with correlation with the learned movement in opto trials ($p = 0.017$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.174). Pre-cue: N = 313/15/11; pre-stimulation: N = 184/15/11 (pairs/sessions/mice).



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Initial state population activity predicts movement variability and movement probability. In Fig. 4, population activity distance to learned activity in the pre-cue/stimulation period was smaller in voluntary trials compared to opto trials. This was because the population activity was measured in the principal component space defined by all voluntary trials. The current figure panels a-c are equivalent to Fig. 4 except that PCA was performed using a random subset (50 %) of only the learned movement trials ("learned activity"). a. Population activity distance to the learned activity before the trial start negatively correlated with correlation with the learned movement in opto trials (left, $p = 0.016$, mixed-effects model, activity distance; Pearson's correlation, R = −0.177), and in voluntary trials ($p = 2.54 \times 10^{-6}$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.262). Each dot represents an individual trial. Opto: N = 184/15/11; voluntary: N = 313/15/11 (pairs/sessions/mice). b. Population activity distance to the learned activity before the trial start showed a correlation between pre- and post-stimulation periods in opto trials (left, $p = 1.27 \times 10^{-6}$ for pre-stimulation activity distance, mixed-effects model; Pearson's correlation, R = 0.834) and between pre- and post-cue periods in voluntary trials (right, $p = 3.51 \times 10^{-4}$ for pre-cue activity distance, mixed-effects model; Pearson's correlation, R = 0.850). Opto: N = 184/15/11; voluntary: N = 313/15/11 (pairs/sessions/mice). c. Population activity distance before the trial start negatively correlated with correlation of subsequent movements between voluntary trials (left, $p = 9.40 \times 10^{-5}$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.091), between opto trials (middle, $p = 0.003$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.103), and between voluntary and opto trials (right, $p = 1.07 \times 10^{-4}$ for activity distance, mixed-effects model; Pearson's correlation, R = −0.083). Voluntary: N = 3413/15/11; opto: N = 1173/15/11; voluntary-opto: N = 4131/15/11 (pairs/sessions/mice). d. A Lasso logistic decoder trained on pre-cue population activity of voluntary trials was applied to opto and omission trials. Lines show separately fit mixed-effects curves relating initial state (decoder output) to movement probability. Initial state predicts movement probability in both opto and omission trials (Opto: $p = 2.20 \times 10^{-5}$, omission: $p = 0.011$ for initial state, mixed-effects model). However, the two curves significantly differ from each other ($\chi^2(2) = 15.83$, $p = 3.65 \times 10^{-4}$, likelihood ratio test). Dots and error bars show the mean ± SEM of the actual movement probability and initial state within quartile bins of initial state. Opto: N = 94/6/5; omission: N = 34/6/5 (trials/sessions/mice).



Extended Data Fig. 10 | Population activity in non-target neurons in no-movement trials. a. Average population activity of non-target neurons in early-target stimulation movement trials (left) and no-movement trials (right). Mean ± SEM pooled across sessions. b. Same as a but for late-target stimulation.

nature portfolio

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Last updated by author(s): Aug 31, 2026

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Image acquisition was performed using Prairie View Software (5.8.64.500 - 5.8.64.900 Bruker). Behavioral data as in lever position was recorded using WaveSurfer (v1.0.5 HHMI) on MATLAB (2019a) and Bpod (SanWorks) on MATLAB (2020a). Bpod also detected lever threshold-crossing and controlled behavioral program working with custom code in MATLAB. Imaging and behavioral data were synchronized using WaveSurfer on MATLAB (2019a).
Data analysis	Imaging and behavior data were analyzed using a combination of previously published code from the lab and additional custom code to fit the needs of the current experiment (MATLAB). Calcium image processing was performed using suite2p (v0.14). Matching of individual neurons across pre-opto and opto sessions was performed using ROIMatchPub (https://github.com/ransona/ROIMatchPub) for automatic detection, followed by manual matching with custom code (MATLAB).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

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All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
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- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Data and code are deposited to Zenodo, available at 10.5281/zenodo.22216381

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No formal statistical tests were used to pre-determine sample sizes. However, sample sizes are comparable to those reported in previous publications. https://doi.org/10.1038/s41586-025-08962-8
Data exclusions	Mice were excluded if they did not achieve expert-level performance over learning.
Replication	Experiments were performed across multiple independent cohorts of animals, with the specific sample sizes detailed for each analysis in the corresponding figure legends. Additionally, two experimenters independently conducted these procedures and observed consistent effects across cohorts.
Randomization	Animals used in this study were not selected based on any prerequisite features other than general well-being. Animals were randomly assigned to experimental groups. Whenever possible, different experimental conditions were repeated in the same animals.
Blinding	Experimenters were not blinded to group assignments during data collection because knowledge of the condition was required to execute the experimental procedures. However, both imaging and behavioral data were processed using automated, unbiased analysis pipelines, with identical inclusion and exclusion criteria applied across conditions. Furthermore, because the primary analytical parameters were computed through multiple downstream processing steps, it is highly unlikely that knowledge of the experimental conditions could introduce bias into the findings.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Mice - C57BL/6 (Charles River). The experiments were done on young adults (8 – 15 weeks old).
Wild animals	No wild animals were used in this study.
Reporting on sex	Both males and females were used in this study.
Field-collected samples	No field-collected samples were used in this study.
Ethics oversight	All procedures were conducted in compliance with protocols approved by the UCSD Institutional Animal Care and Use Committee (IACUC) and guidelines of the National Institutes of Health (NIH).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A